johnston-linear-matrix-algebra: image evidence

Image regions

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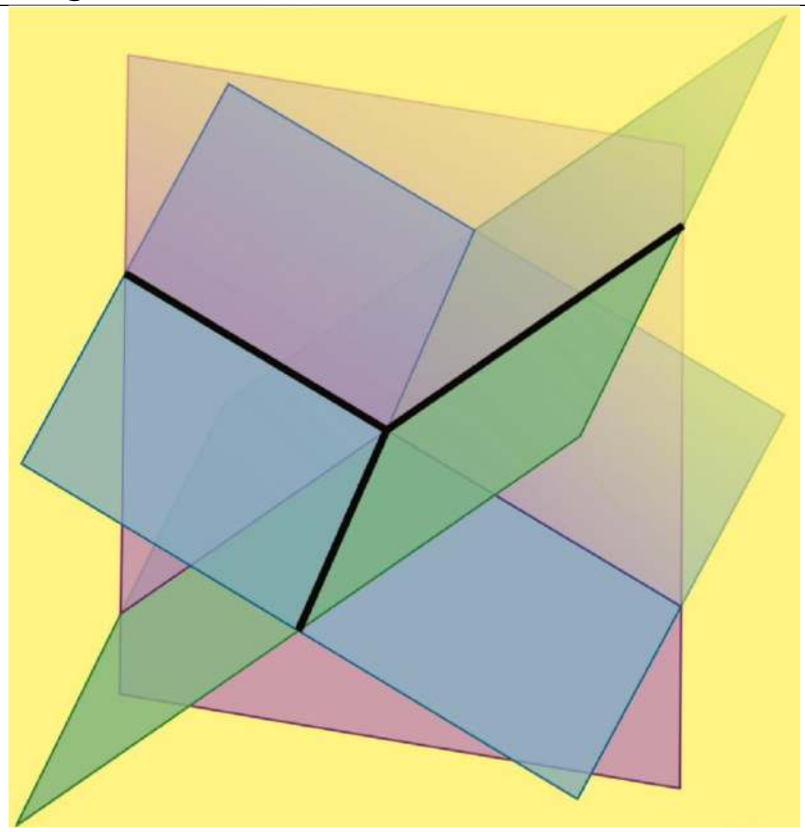
johnston-linear-matrix-algebra_DIA_0001	1	—	—	—	

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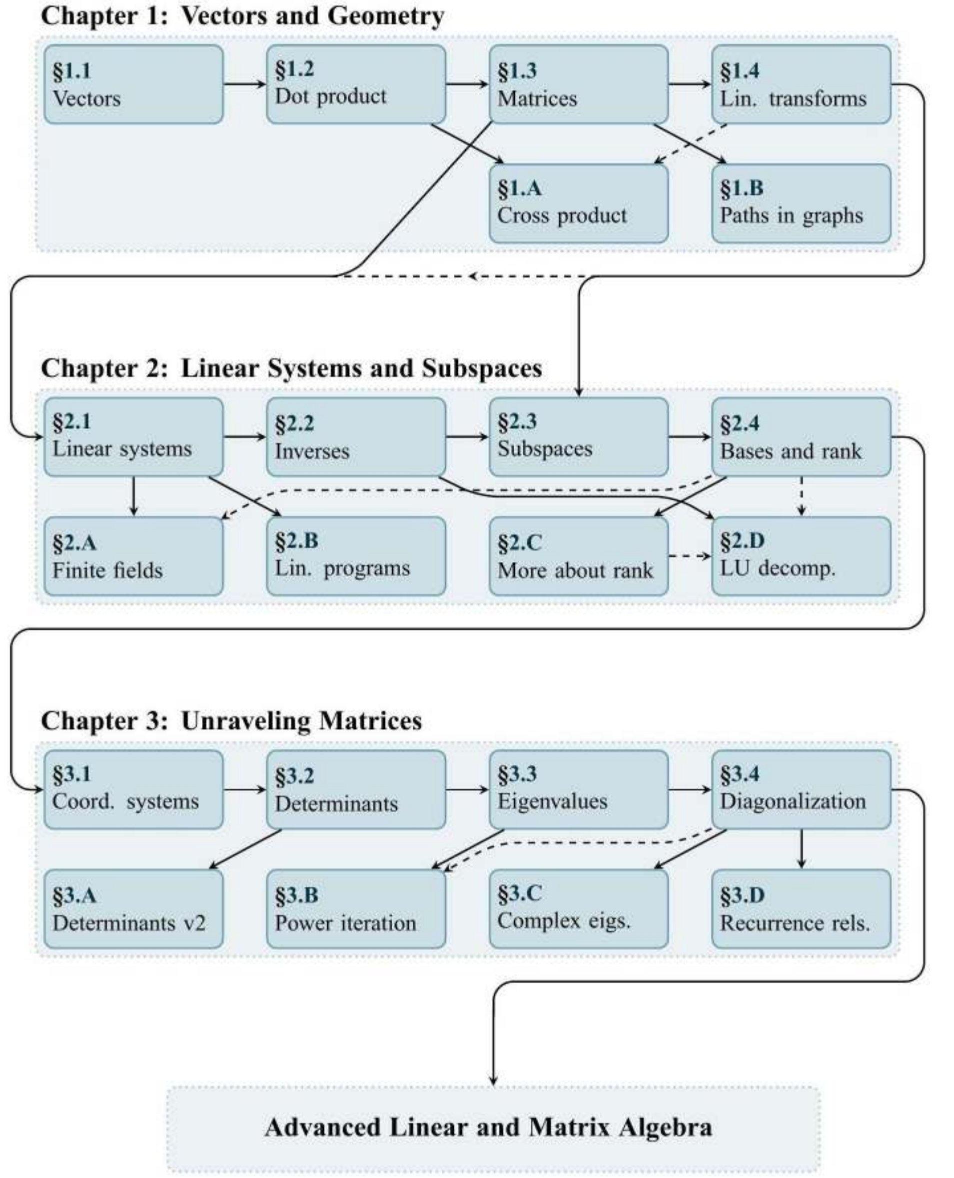

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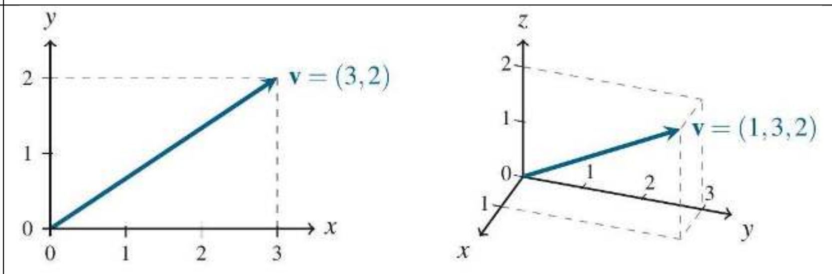

(a) The vector $\mathbf{v} = (3,2) \in \mathbb{R}^2$. (b) The vector $\mathbf{v} = (1,3,2) \in \mathbb{R}^3$.
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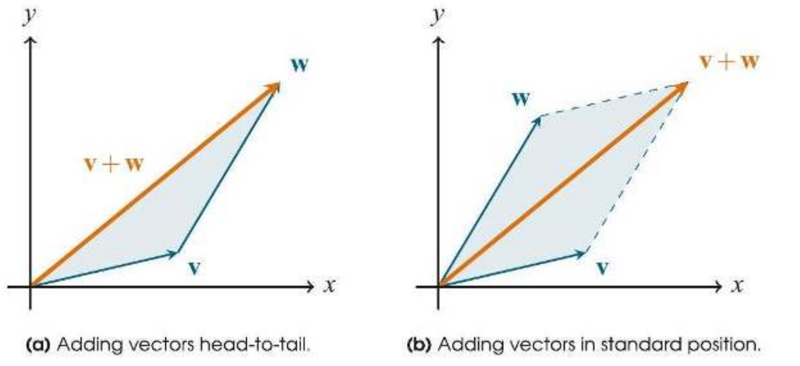
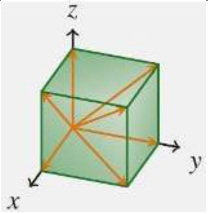
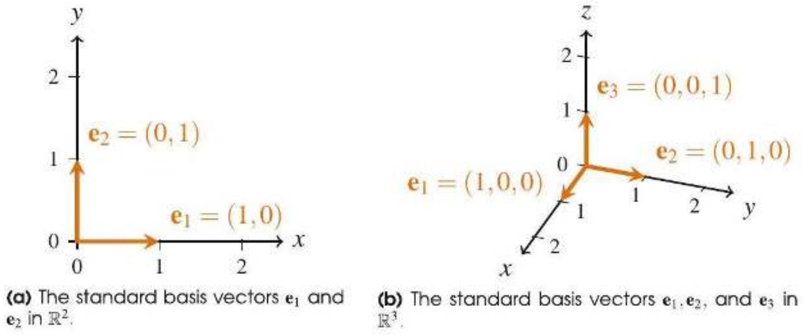
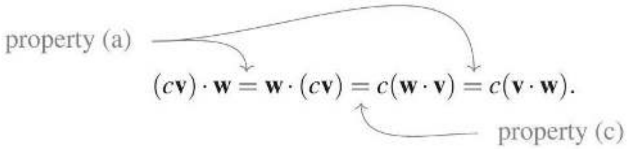
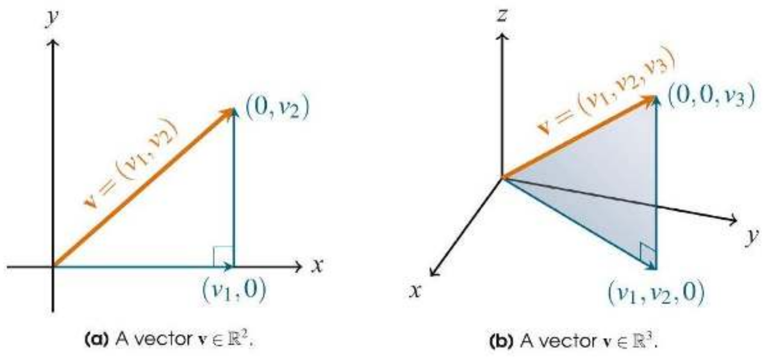
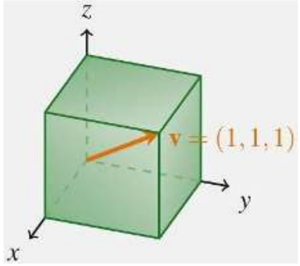
johnston-linear-matrix-algebra_DIA_0004	18	—	—	—	
johnston-linear-matrix-algebra_DIA_0005	19	—	—	—	
johnston-linear-matrix-algebra_DIA_0006	23	—	—	—	
johnston-linear-matrix-algebra_DIA_0007	27	—	—	—	
johnston-linear-matrix-algebra_DIA_0008	28	—	—	—	
johnston-linear-matrix-algebra_DIA_0009	29	—	—	—	

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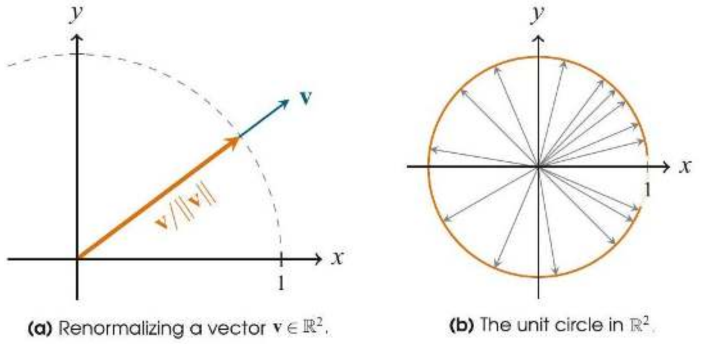
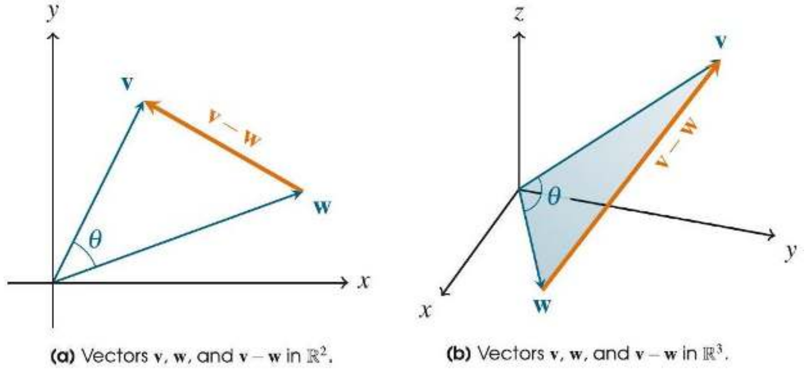
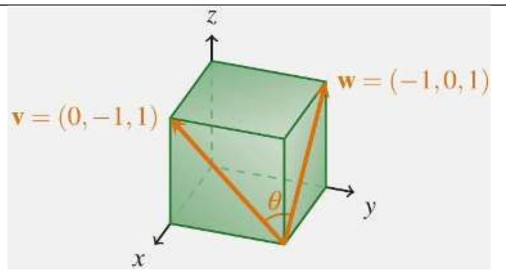
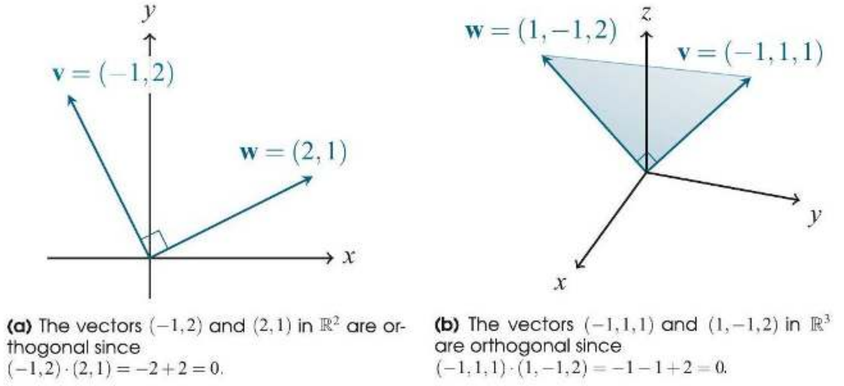
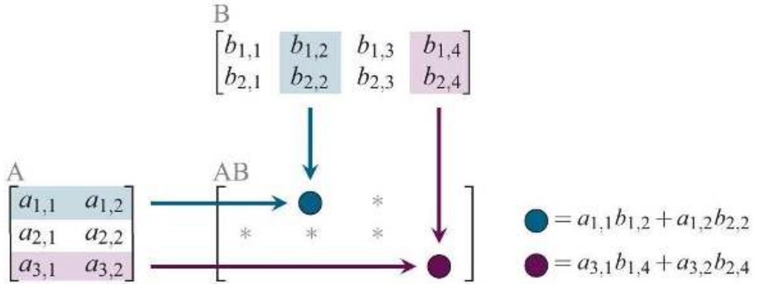
johnston-linear-matrix-algebra_DIA_0010	30	—	—	—	
johnston-linear-matrix-algebra_DIA_0011	32	—	—	—	
johnston-linear-matrix-algebra_DIA_0012	33	—	—	—	
johnston-linear-matrix-algebra_DIA_0013	34	—	—	—	
johnston-linear-matrix-algebra_DIA_0014	38	—	—	—	

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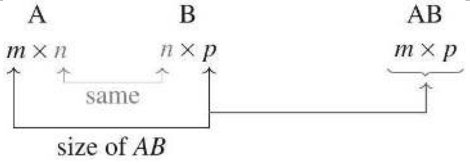
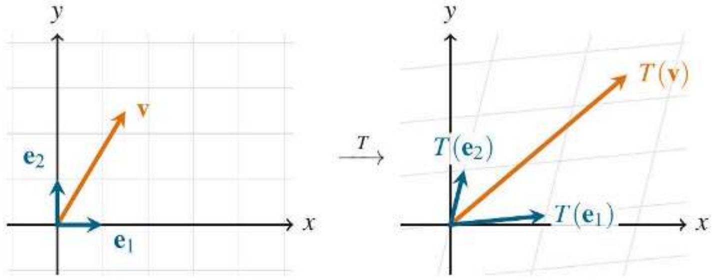
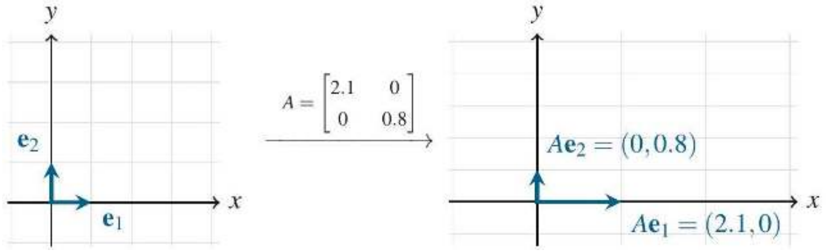
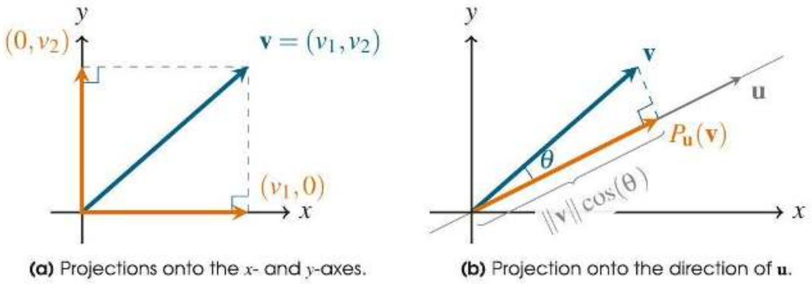
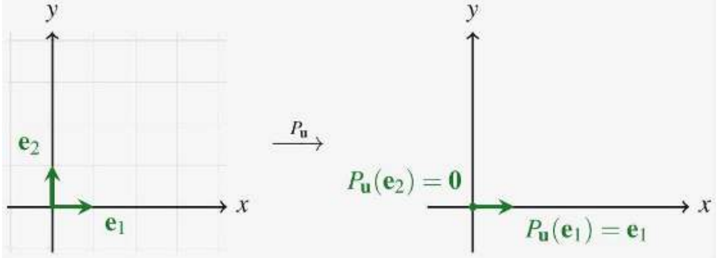
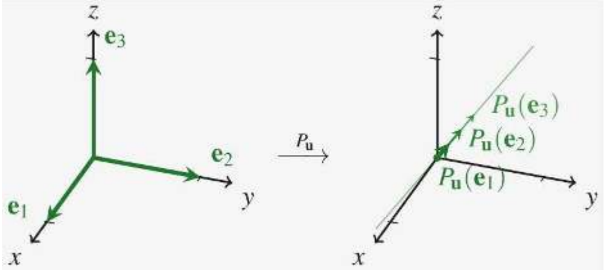
johnston-linear-matrix-algebra_DIA_0015	39	—	—	—	
johnston-linear-matrix-algebra_DIA_0016	51	—	—	—	
johnston-linear-matrix-algebra_DIA_0017	57	—	—	—	
johnston-linear-matrix-algebra_DIA_0018	58	—	—	—	
johnston-linear-matrix-algebra_DIA_0019	59	—	—	—	
johnston-linear-matrix-algebra_DIA_0020	59	—	—	—	

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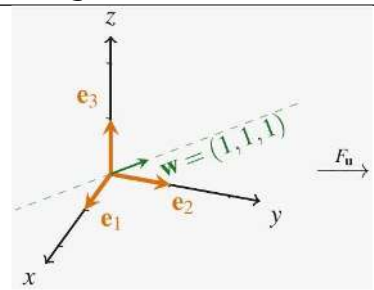
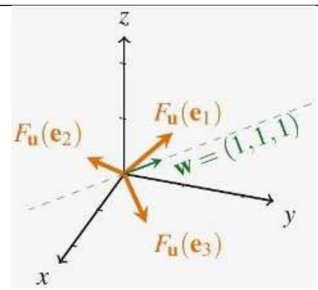
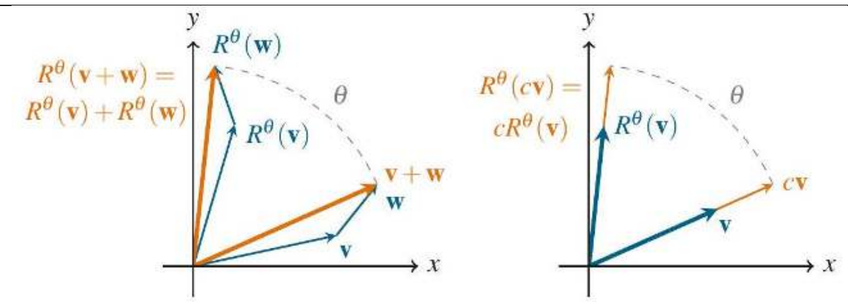
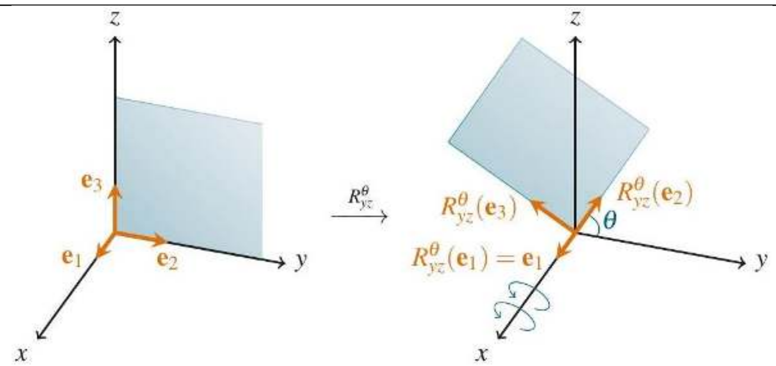
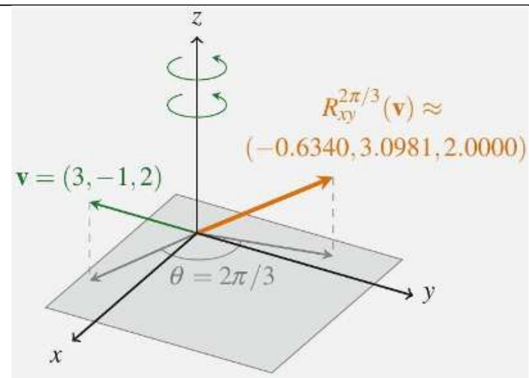
johnston-linear-matrix-algebra_DIA_0021	61	—	—	—	
johnston-linear-matrix-algebra_DIA_0022	61	—	—	—	
johnston-linear-matrix-algebra_DIA_0023	62	—	—	—	**(a)** Rotating $\mathbf{v}$ and $\mathbf{w}$ and then adding them has the same effect as adding and then rotating them, so $R^\theta(\mathbf{v}+\mathbf{w}) = R^\theta(\mathbf{v}) + R^\theta(\mathbf{w})$. **(b)** Rotating $\mathbf{v}$ and then multiplying it by c has the same effect as multiplying by c and then rotating, so $R^\theta(c\mathbf{v}) = cR^\theta(\mathbf{v})$.
johnston-linear-matrix-algebra_DIA_0024	65	—	—	—	
johnston-linear-matrix-algebra_DIA_0025	66	—	—	—	

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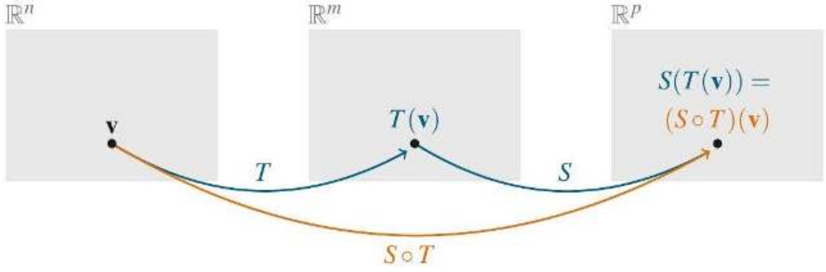
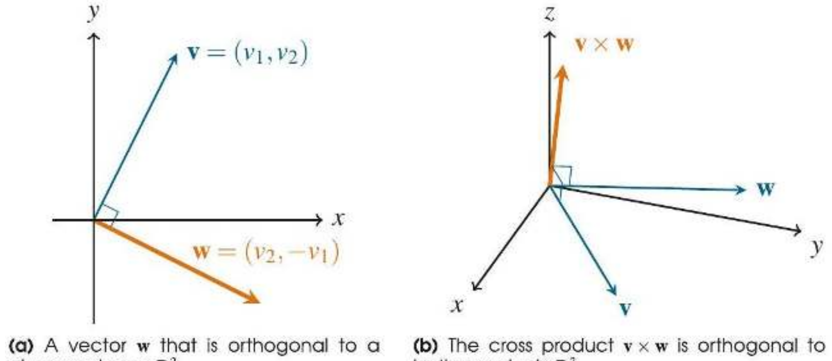
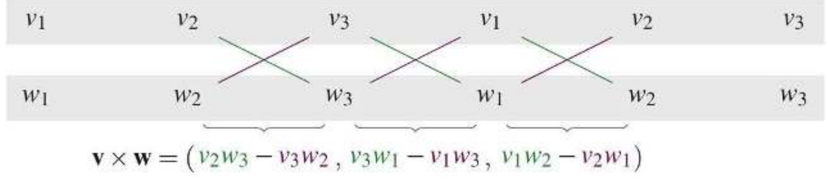
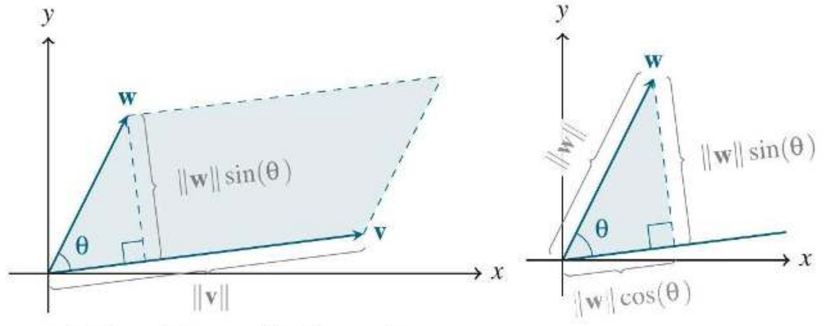
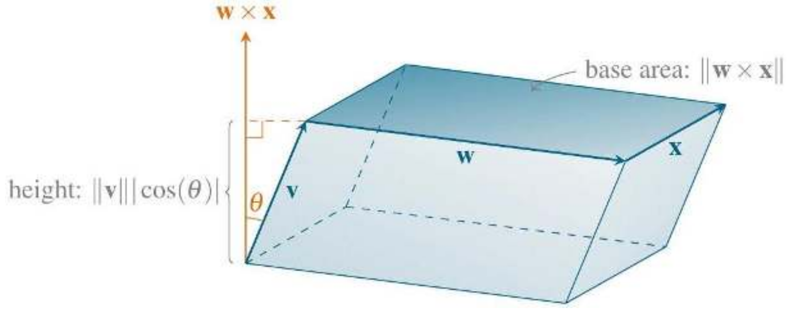
johnston-linear-matrix-algebra_DIA_0026	67	—	—	—	
johnston-linear-matrix-algebra_DIA_0027	74	—	—	—	 (a) A vector $\mathbf{w}$ that is orthogonal to a given vector $\mathbf{v} \in \mathbb{R}^2$. (b) The cross product $\mathbf{v} \times \mathbf{w}$ is orthogonal to both $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^3$.
johnston-linear-matrix-algebra_DIA_0028	74	—	—	—	
johnston-linear-matrix-algebra_DIA_0029	76	—	—	—	 (a) A parallelogram with sides $\mathbf{v}$ and $\mathbf{w}$. (b) The triangle formed by $\mathbf{w}$.
johnston-linear-matrix-algebra_DIA_0030	78	—	—	—	

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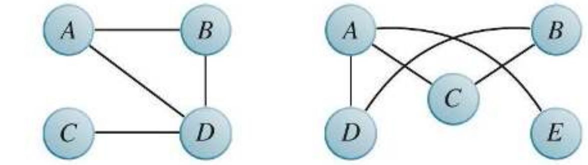
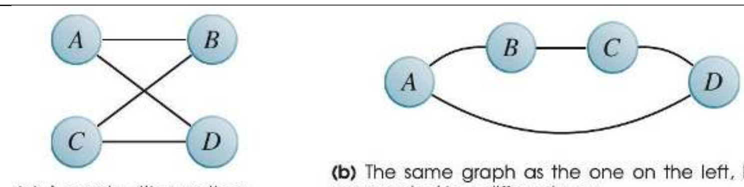
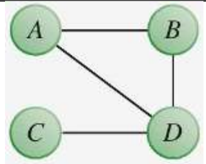
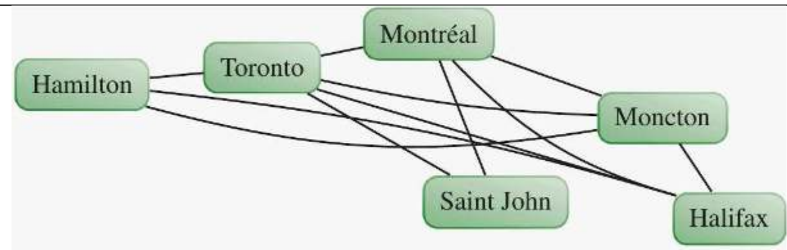
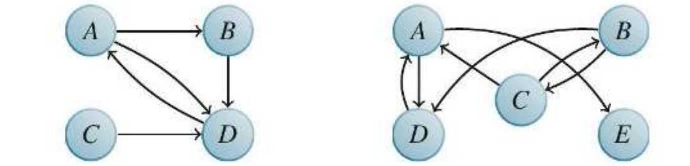
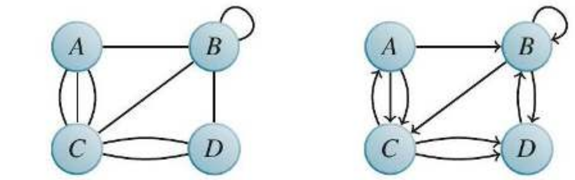
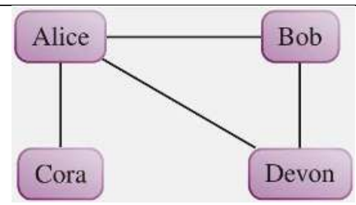
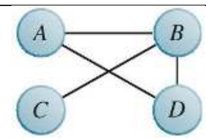
johnston-linear-matrix-algebra_DIA_0031	80	—	—	—	 (a) A graph with 4 vertices. (b) A graph with 5 vertices.
johnston-linear-matrix-algebra_DIA_0032	81	—	—	—	 (a) A graph with 4 vertices. (b) The same graph as the one on the left, but represented in a different way.
johnston-linear-matrix-algebra_DIA_0033	83	—	—	—	
johnston-linear-matrix-algebra_DIA_0034	84	—	—	—	
johnston-linear-matrix-algebra_DIA_0035	86	—	—	—	 (a) A directed graph with 4 vertices. (b) A directed graph with 5 vertices.
johnston-linear-matrix-algebra_DIA_0036	87	—	—	—	 (a) An undirected multigraph. (b) A directed multigraph.
johnston-linear-matrix-algebra_DIA_0037	88	—	—	—	
johnston-linear-matrix-algebra_DIA_0038	90	—	—	—	

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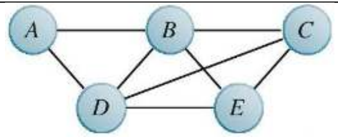
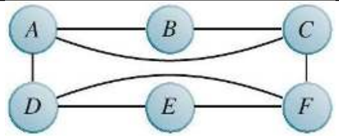
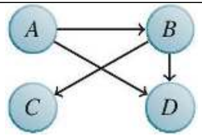
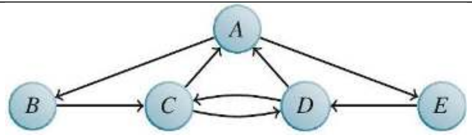
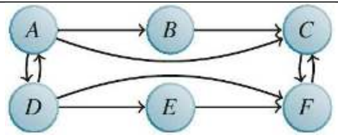
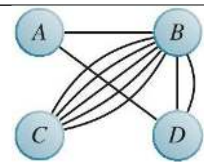
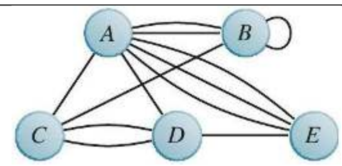
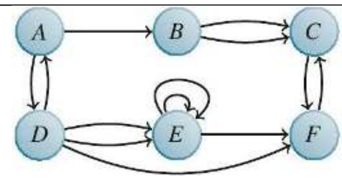
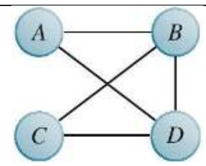
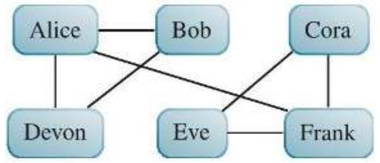
johnston-linear-matrix-algebra_DIA_0039	90	—	—	—	
johnston-linear-matrix-algebra_DIA_0040	90	—	—	—	
johnston-linear-matrix-algebra_DIA_0041	90	—	—	—	
johnston-linear-matrix-algebra_DIA_0042	90	—	—	—	
johnston-linear-matrix-algebra_DIA_0043	90	—	—	—	
johnston-linear-matrix-algebra_DIA_0044	90	—	—	—	
johnston-linear-matrix-algebra_DIA_0045	90	—	—	—	
johnston-linear-matrix-algebra_DIA_0046	90	—	—	—	
johnston-linear-matrix-algebra_DIA_0047	90	—	—	—	
johnston-linear-matrix-algebra_DIA_0048	91	—	—	—	

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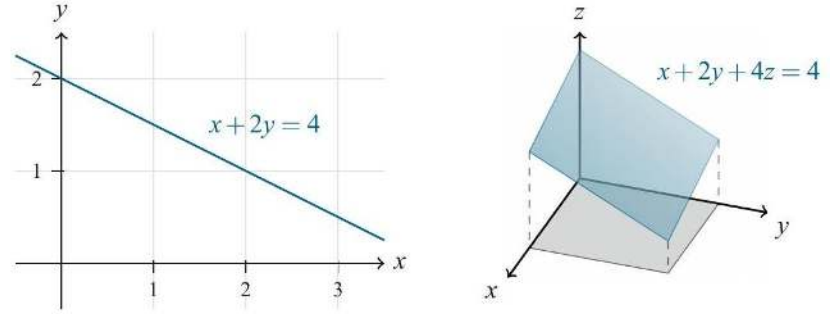
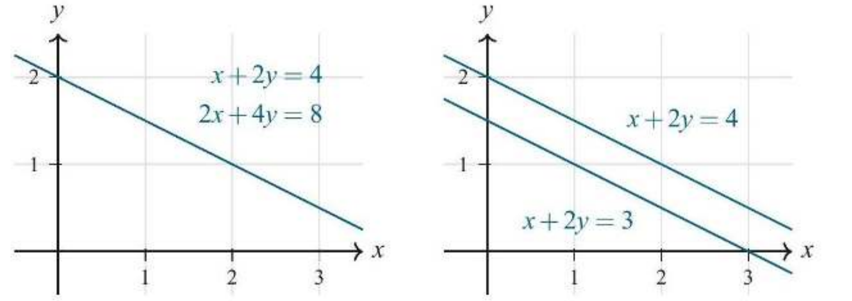
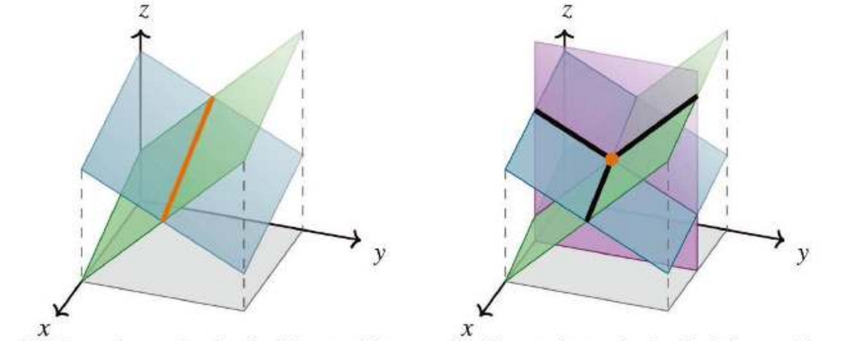
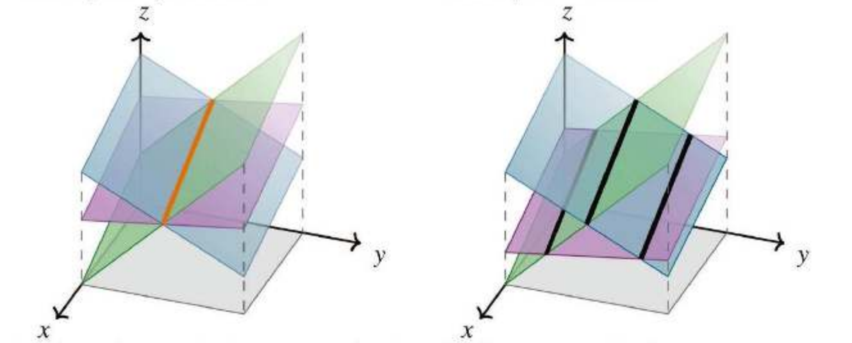
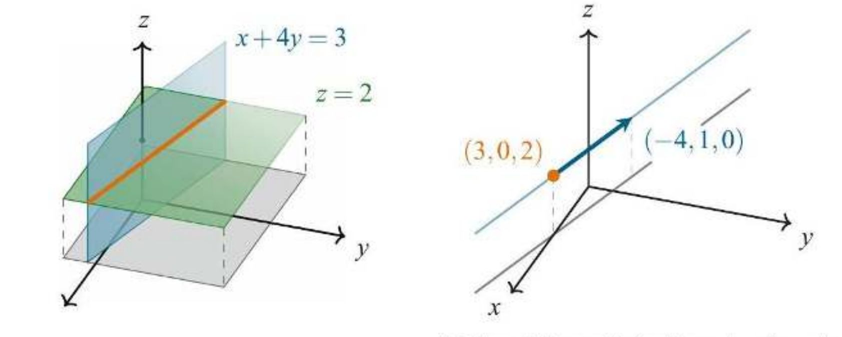
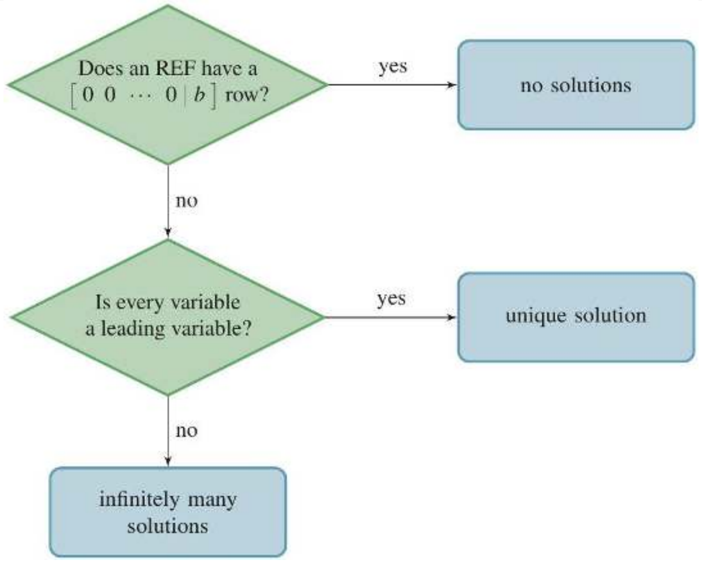
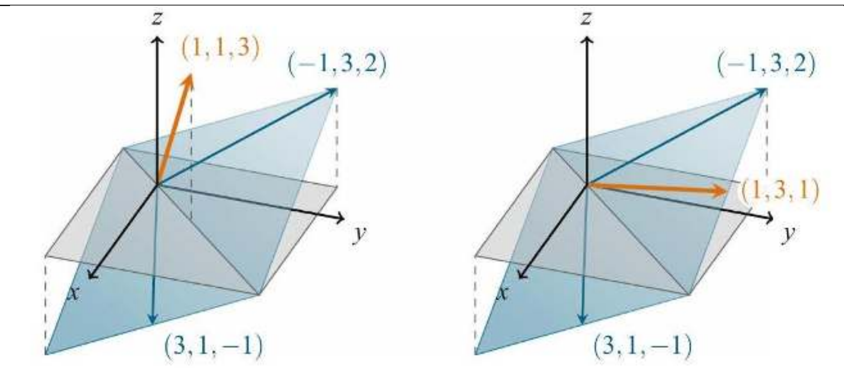
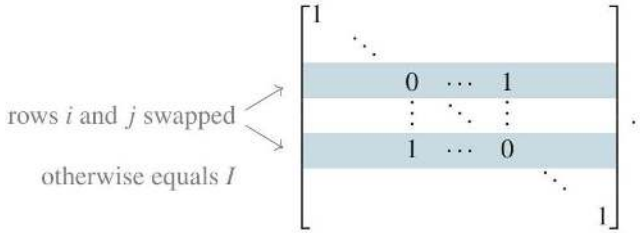
johnston-linear-matrix-algebra_DIA_0049	93	—	—	—	 **(a)** A linear equation in 2 variables represents a line in $\mathbb{R}^2$. The equation of this line is perhaps more usually written in the (equivalent) form $y=-x/2+2$. **(b)** A linear equation in 3 variables represents a plane in $\mathbb{R}^3$. This plane actually extends arbitrarily far in all directions—it is just truncated here so it is easier to visualize.
johnston-linear-matrix-algebra_DIA_0050	94	—	—	—	 **(a)** A linear system with infinitely many solutions, represented by two equivalent (and thus always intersecting) lines. **(b)** A linear system with no solutions, represented by two parallel (and thus non-intersecting) lines.
johnston-linear-matrix-algebra_DIA_0051	97	—	—	—	 **(a)** Two planes typically intersect in a line, and thus a system of two linear equations in three variables usually has infinitely many solutions. **(b)** Three planes typically intersect in a point, and thus a system of three linear equations in three variables usually has exactly one solution.  **(c)** Three (or more) planes can also intersect in a line, and thus a system of three or more linear equations in three variables can have infinitely many solutions. **(d)** Three (or more) planes can also not have a common intersection at all, and thus a system of three or more linear equations in three variables can have no solutions.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0052	106	—	—	—	 **(a)** The solution set is the intersection of the two planes $x+4y=3$ and $z=2$. **(b)** The solution set is the line going through the point $(3,0,2)$ in the direction of the vector $(-4,1,0)$.
johnston-linear-matrix-algebra_DIA_0053	106	—	—	—	
johnston-linear-matrix-algebra_DIA_0054	111	—	—	—	 **(a)** $(1,1,3)$ is not a linear combination of $(-1,3,2)$ and $(3,1,-1)$. **(b)** $(1,3,1)$ is a linear combination of $(-1,3,2)$ and $(3,1,-1)$.
johnston-linear-matrix-algebra_DIA_0055	117	—	—	—	

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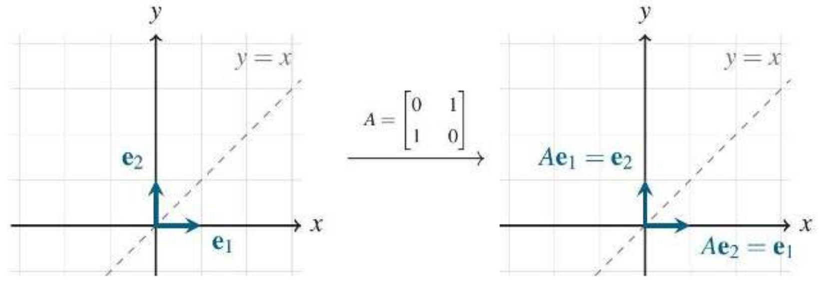
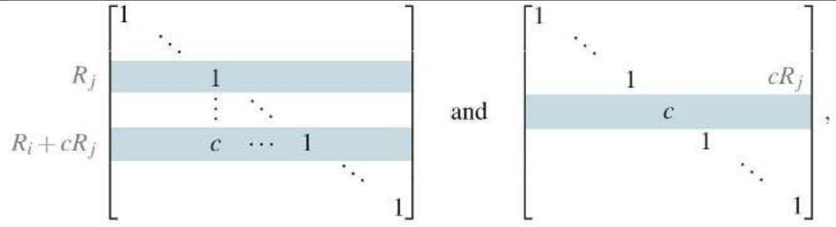
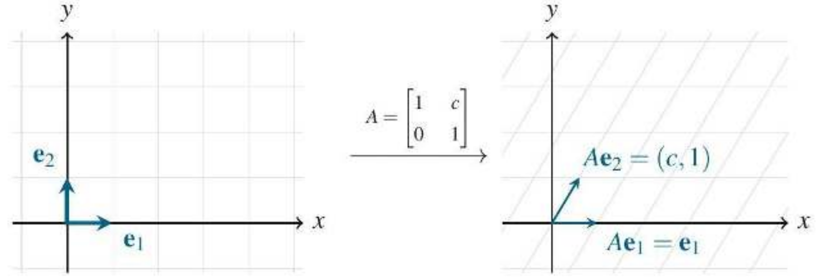
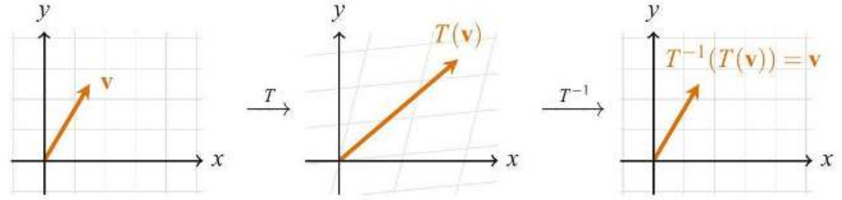
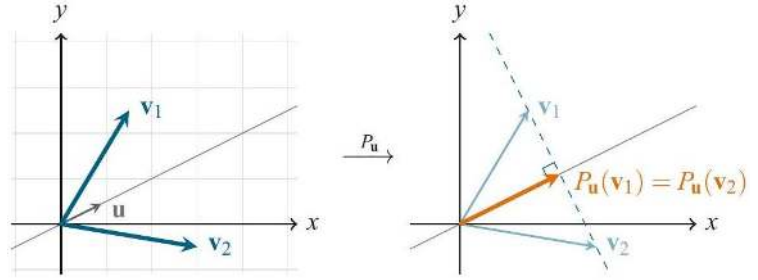
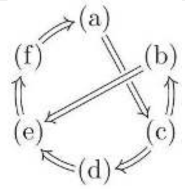
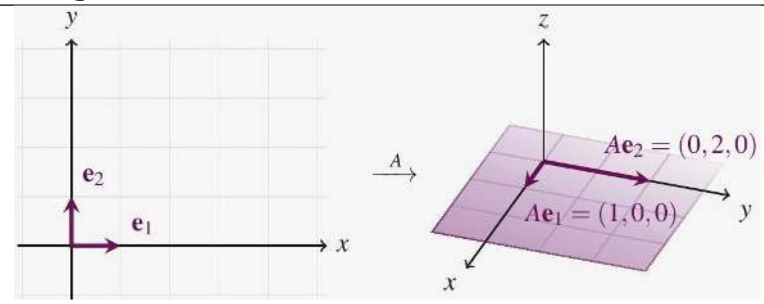
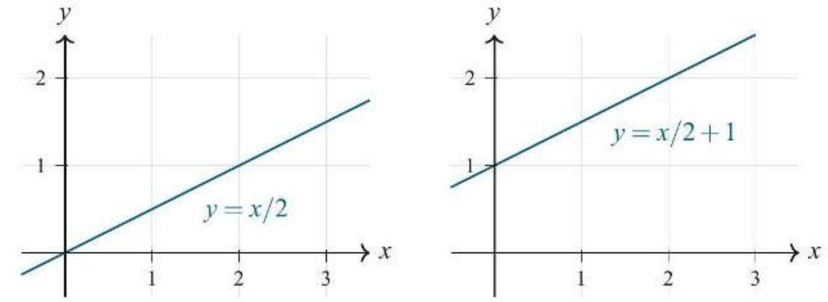
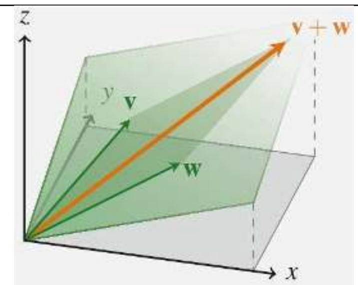
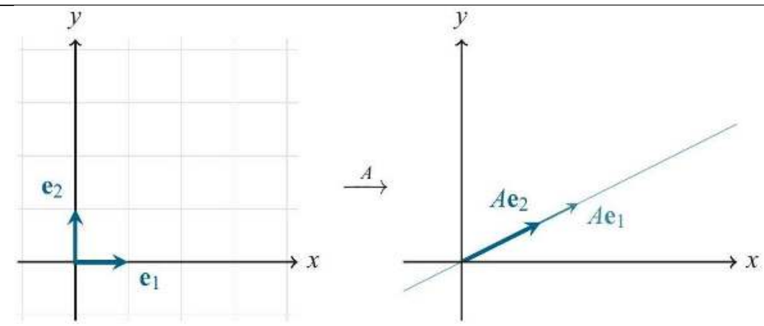
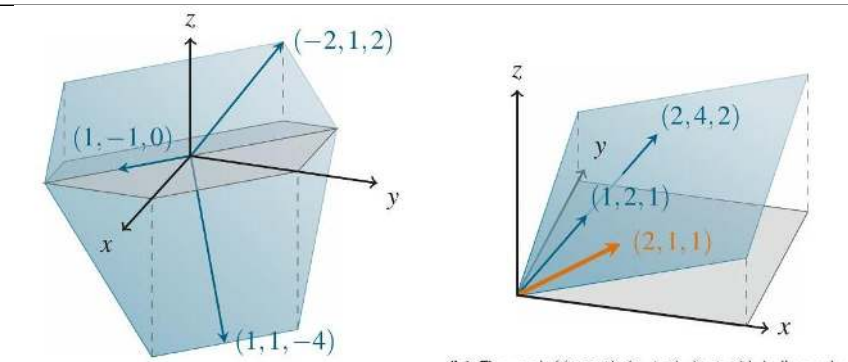
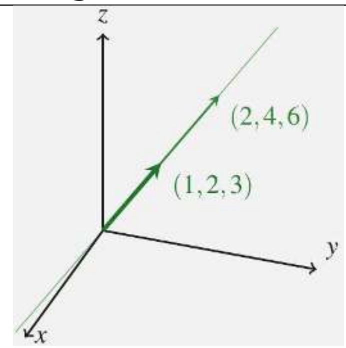
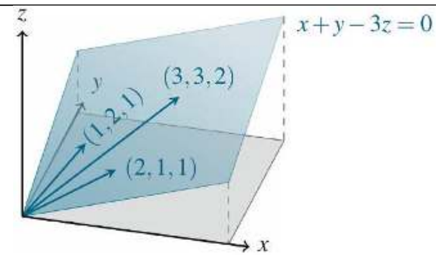
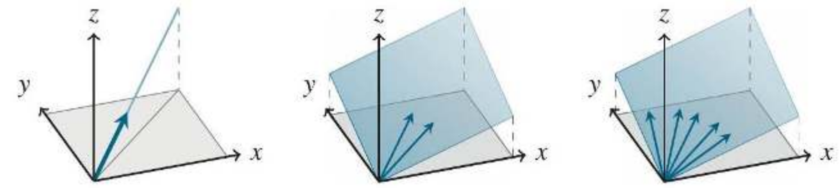
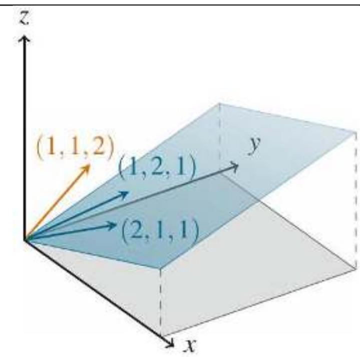
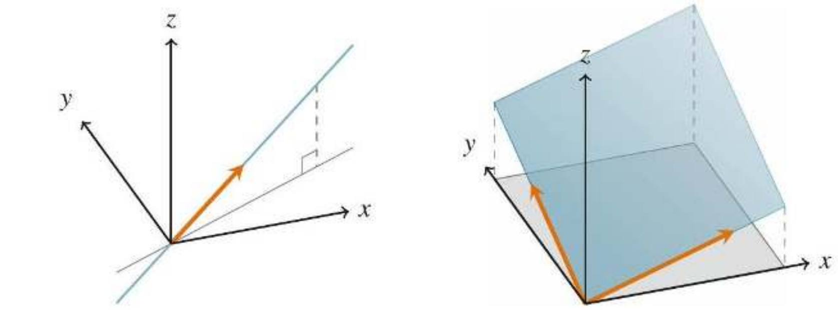
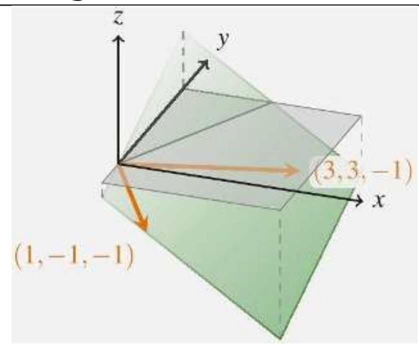
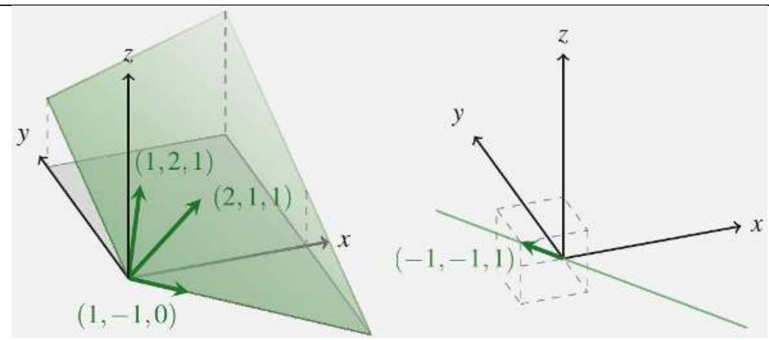
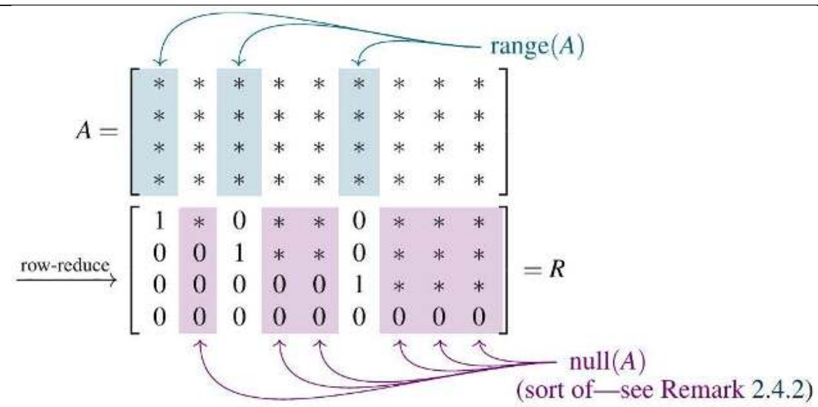
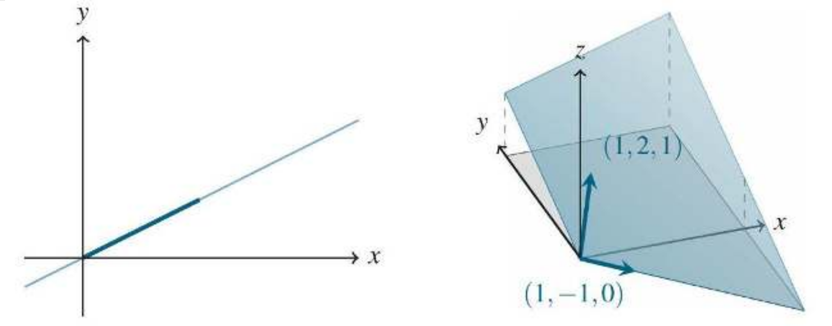
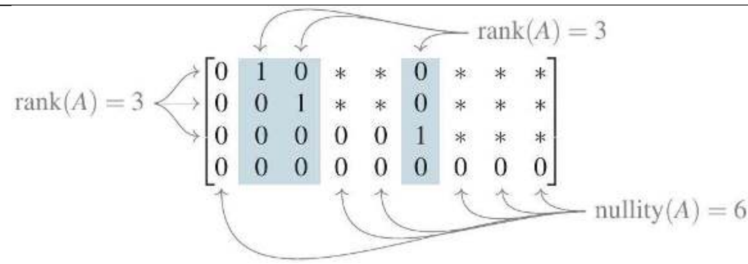
johnston-linear-matrix-algebra_DIA_0056	118	—	—	—	
johnston-linear-matrix-algebra_DIA_0057	118	—	—	—	
johnston-linear-matrix-algebra_DIA_0058	118	—	—	—	
johnston-linear-matrix-algebra_DIA_0059	123	—	—	—	
johnston-linear-matrix-algebra_DIA_0060	124	—	—	—	
johnston-linear-matrix-algebra_DIA_0061	127	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0062	133	—	—	—	
johnston-linear-matrix-algebra_DIA_0063	137	—	—	—	 **(a)** The line $y = x/2$ is a subspace, since lines through the origin are subspaces. **(b)** The line $y = x/2+1$ is not a subspace, since it does not go through the origin.
johnston-linear-matrix-algebra_DIA_0064	138	—	—	—	
johnston-linear-matrix-algebra_DIA_0065	145	—	—	—	
johnston-linear-matrix-algebra_DIA_0066	149	—	—	—	 **(a)** A linearly dependent set of 3 vectors that spans a plane in $\mathbb{R}^3$. **(b)** The set $\{(1,2,1),(2,4,2),(2,1,1)\}$ is linearly dependent, but removing $(2,1,1)$ from it changes its span from a plane to a line.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0067	150	—	—	—	
johnston-linear-matrix-algebra_DIA_0068	155	—	—	—	
johnston-linear-matrix-algebra_DIA_0069	156	—	—	—	**(a)** A single vector spans a line. **(b)** Two non-parallel vectors span a plane. **(c)** Several vectors on a plane span that plane.
johnston-linear-matrix-algebra_DIA_0070	158	—	—	—	
johnston-linear-matrix-algebra_DIA_0071	159	—	—	—	**(a)** A basis of a line contains just 1 vector, so lines are 1-dimensional. **(b)** A basis of a plane contains 2 vectors, so planes are 2-dimensional.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0072	161	—	—	—	
johnston-linear-matrix-algebra_DIA_0073	165	—	—	—	
johnston-linear-matrix-algebra_DIA_0074	166	—	—	—	
johnston-linear-matrix-algebra_DIA_0075	171	—	—	—	**(a)** A projection onto a line has rank 1, since its range is 1-dimensional. **(b)** A 3 × 3 matrix with rank 2 has a plane as its range.
johnston-linear-matrix-algebra_DIA_0076	174	—	—	—	

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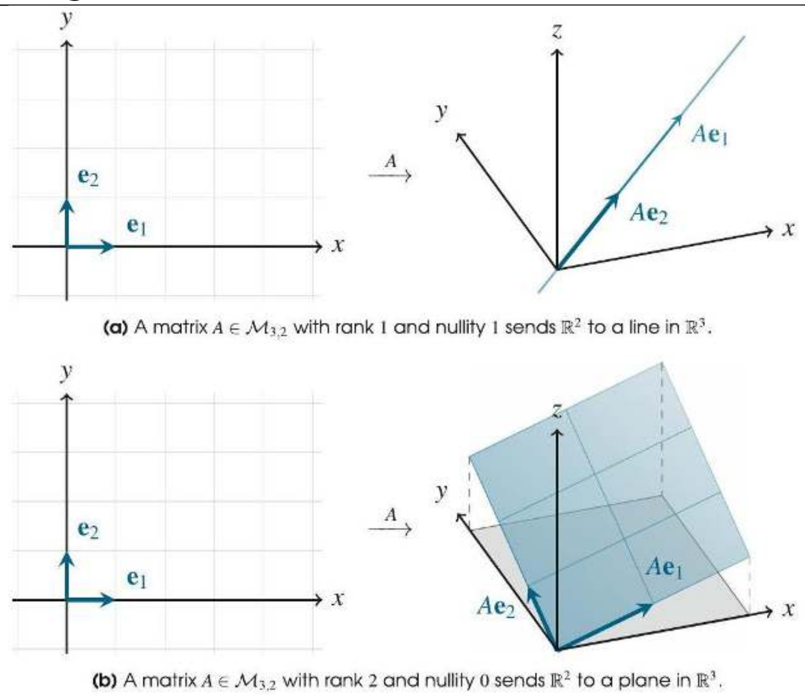
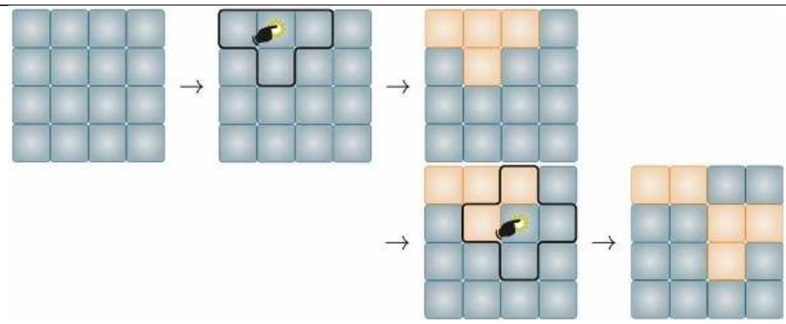
johnston-linear-matrix-algebra_DIA_0077	175	—	—	—	
johnston-linear-matrix-algebra_DIA_0078	184	—	—	—	
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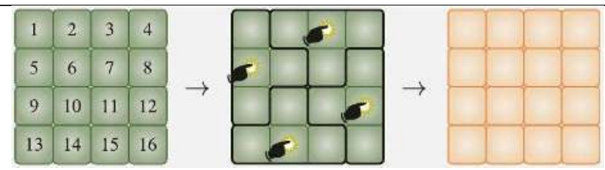
johnston-linear-matrix-algebra_DIA_0080	186	—	—	—	

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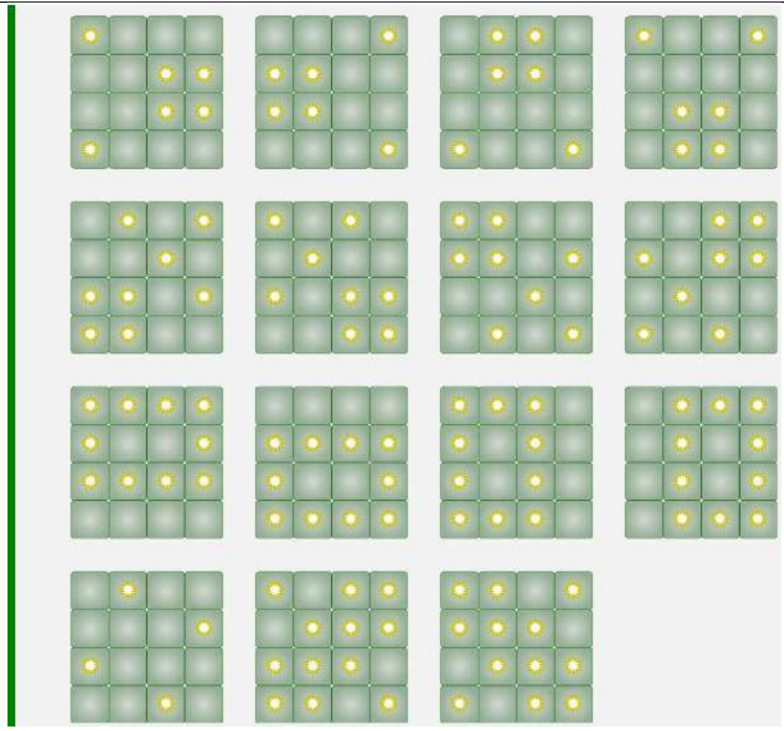
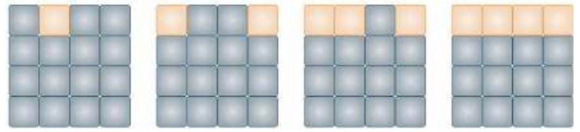
johnston-linear-matrix-algebra_DIA_0081	187	—	—	—	
johnston-linear-matrix-algebra_DIA_0082	187	—	—	—	
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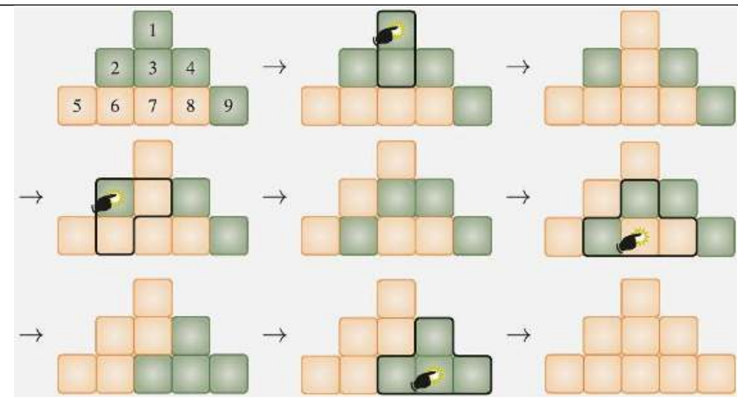
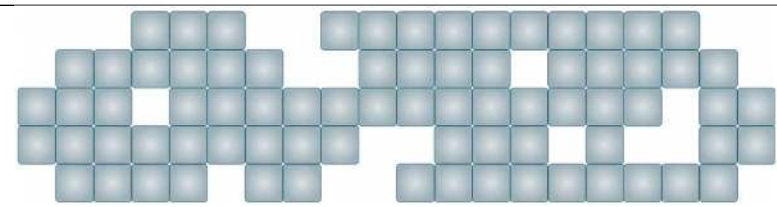
johnston-linear-matrix-algebra_DIA_0084	188	—	—	—	
johnston-linear-matrix-algebra_DIA_0085	189	—	—	—	

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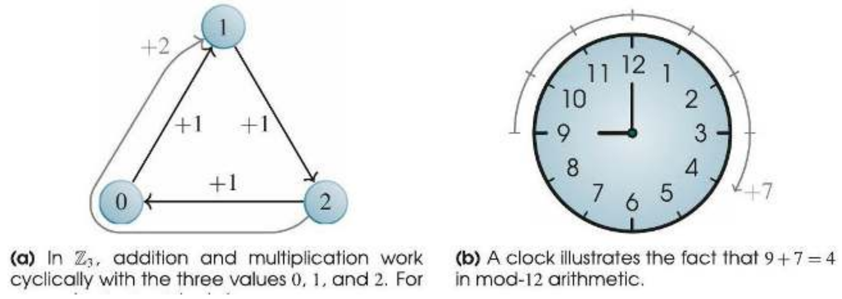
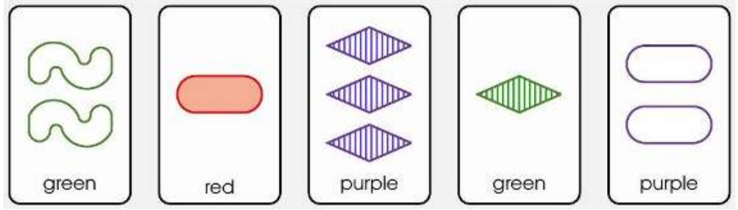
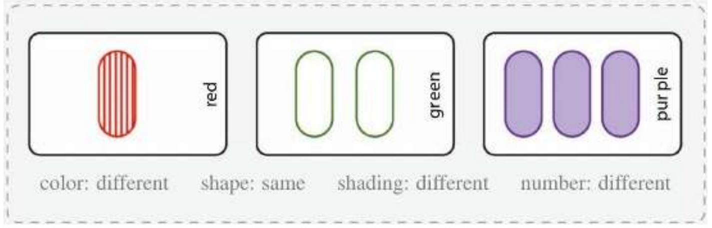
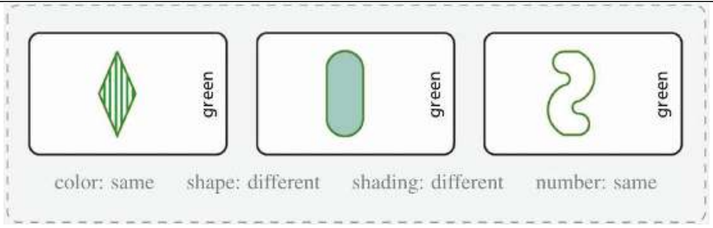
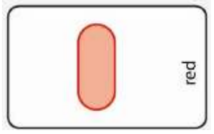
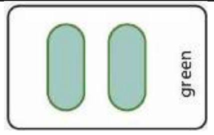
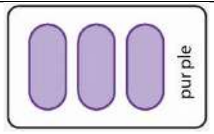
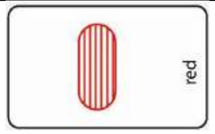
johnston-linear-matrix-algebra_DIA_0086	191	—	—	—	
johnston-linear-matrix-algebra_DIA_0087	192	—	—	—	
johnston-linear-matrix-algebra_DIA_0088	192	—	—	—	
johnston-linear-matrix-algebra_DIA_0089	192	—	—	—	
johnston-linear-matrix-algebra_DIA_0090	194	—	—	—	
johnston-linear-matrix-algebra_DIA_0091	194	—	—	—	
johnston-linear-matrix-algebra_DIA_0092	194	—	—	—	
johnston-linear-matrix-algebra_DIA_0093	194	—	—	—	

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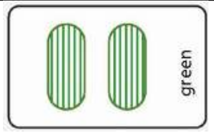
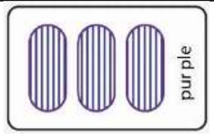
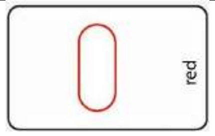
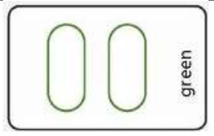
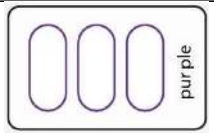
johnston-linear-matrix-algebra_DIA_0094	194	—	—	—	
johnston-linear-matrix-algebra_DIA_0095	194	—	—	—	
johnston-linear-matrix-algebra_DIA_0096	194	—	—	—	
johnston-linear-matrix-algebra_DIA_0097	194	—	—	—	
johnston-linear-matrix-algebra_DIA_0098	194	—	—	—	
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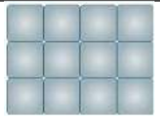
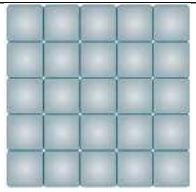
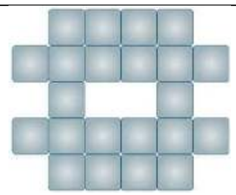
johnston-linear-matrix-algebra_DIA_0100	196	—	—	—	
johnston-linear-matrix-algebra_DIA_0101	196	—	—	—	
johnston-linear-matrix-algebra_DIA_0102	196	—	—	—	
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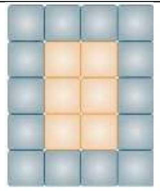
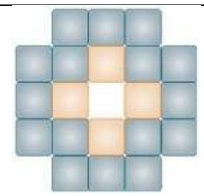
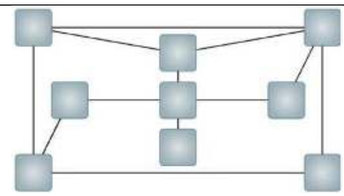
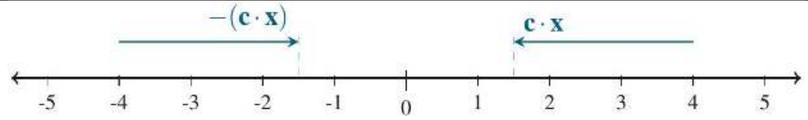
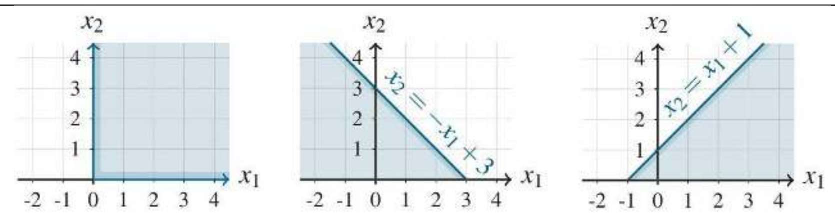
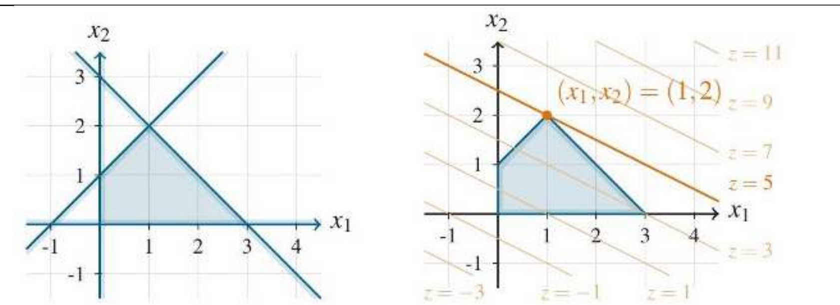
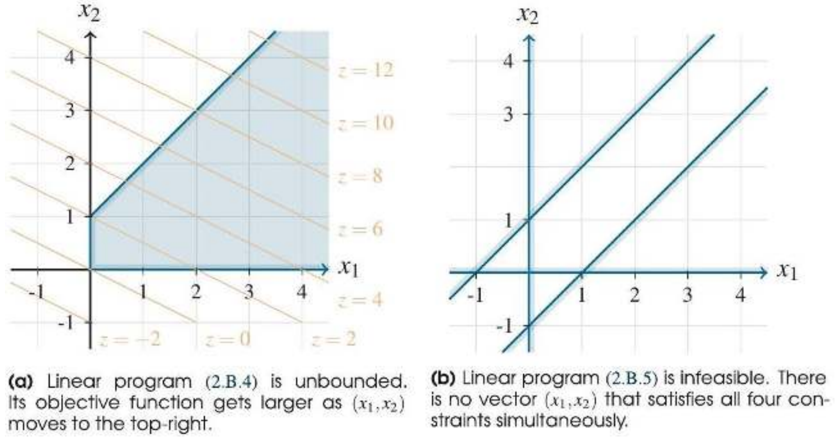
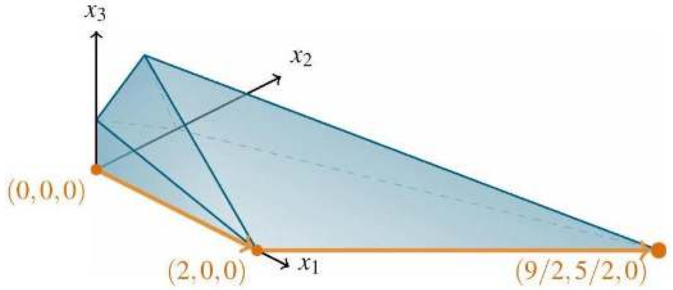
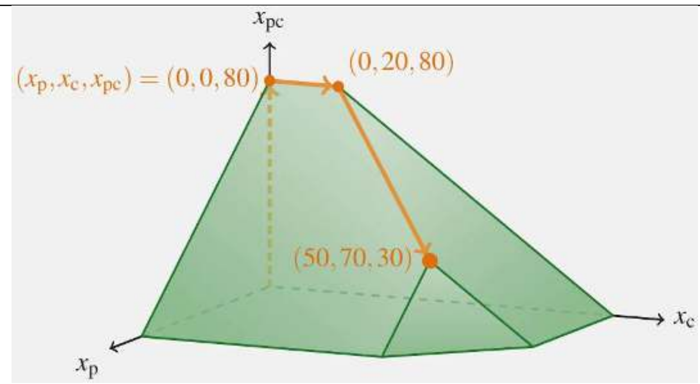
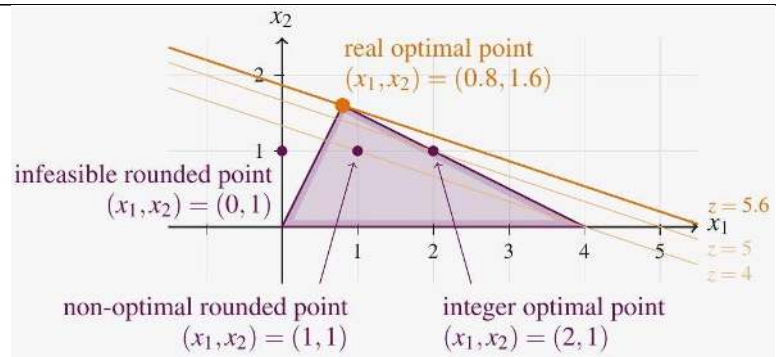
johnston-linear-matrix-algebra_DIA_0105	196	—	—	—	
johnston-linear-matrix-algebra_DIA_0106	196	—	—	—	
johnston-linear-matrix-algebra_DIA_0107	197	—	—	—	
johnston-linear-matrix-algebra_DIA_0108	198	—	—	—	
johnston-linear-matrix-algebra_DIA_0109	202	—	—	—	 **(a)** $x_1, x_2 \geq 0$ **(b)** $x_2 \leq -x_1 + 3$ **(c)** $x_2 \leq x_1 + 1$
johnston-linear-matrix-algebra_DIA_0110	202	—	—	—	 **(a)** The feasible region of a linear program is the intersection of the various half-planes defined by its constraints. **(b)** Of all the lines of the form $x_2 = -x_1/2 + z/2$, the one with largest value of z (i.e., largest y-intercept) that intersects the feasible region is the one with $z = 5$ (i.e., y-intercept $5/2$).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0111	203	—	—	—	
johnston-linear-matrix-algebra_DIA_0112	208	—	—	—	
johnston-linear-matrix-algebra_DIA_0113	210	—	—	—	
johnston-linear-matrix-algebra_DIA_0114	211	—	—	—	

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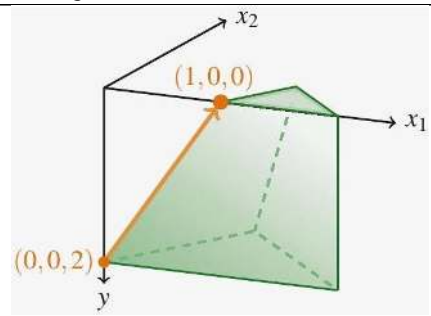
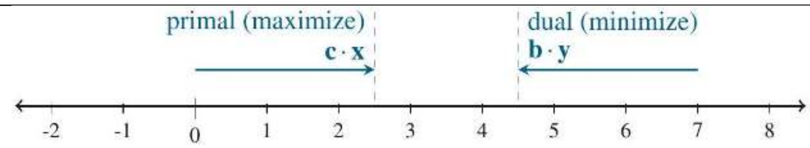
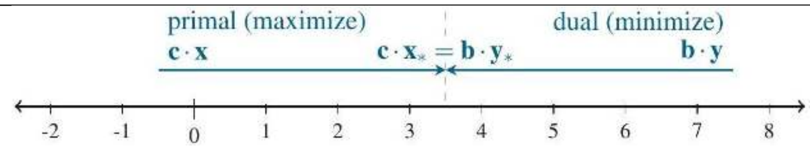
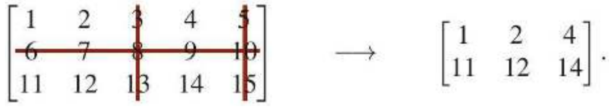
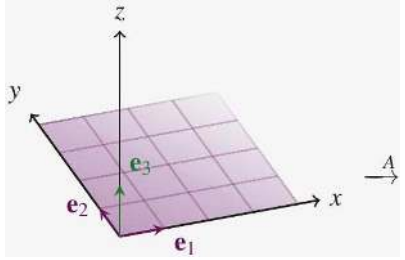
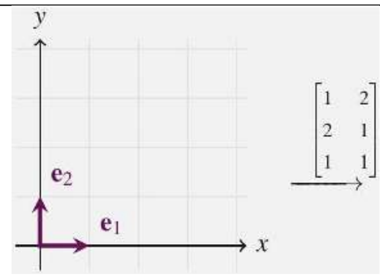
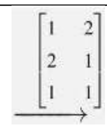
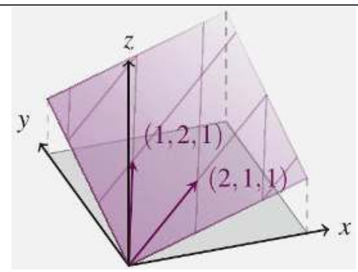
johnston-linear-matrix-algebra_DIA_0115	213	—	—	—	
johnston-linear-matrix-algebra_DIA_0116	216	—	—	—	
johnston-linear-matrix-algebra_DIA_0117	218	—	—	—	
johnston-linear-matrix-algebra_DIA_0118	227	—	—	—	
johnston-linear-matrix-algebra_DIA_0119	228	—	—	—	
johnston-linear-matrix-algebra_DIA_0120	228	—	—	—	
johnston-linear-matrix-algebra_DIA_0121	228	—	—	—	
johnston-linear-matrix-algebra_DIA_0122	228	—	—	—	

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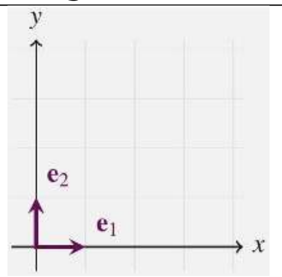
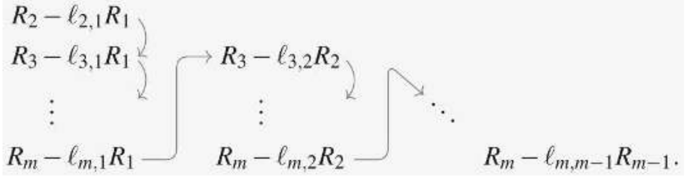
johnston-linear-matrix-algebra_DIA_0123	228	—	—	—	
johnston-linear-matrix-algebra_DIA_0124	234	—	—	—	
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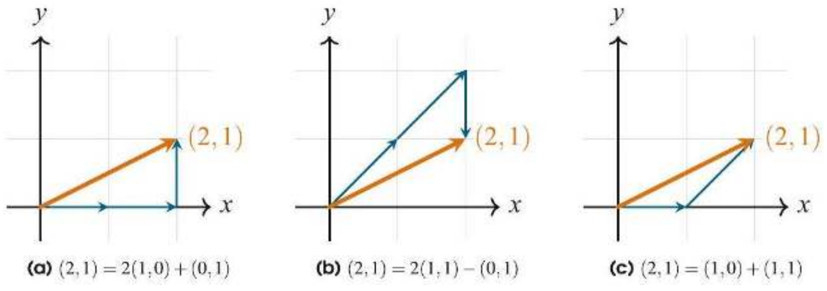
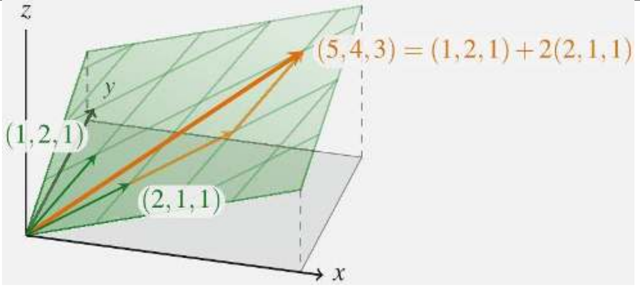
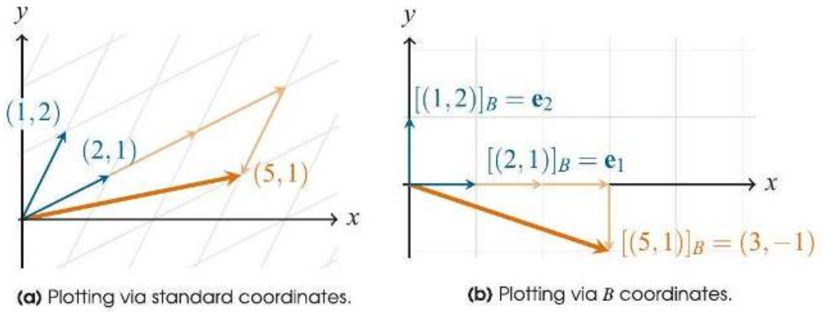
johnston-linear-matrix-algebra_DIA_0126	251	—	—	—	
johnston-linear-matrix-algebra_DIA_0127	253	—	—	—	
johnston-linear-matrix-algebra_DIA_0128	254	—	—	—	

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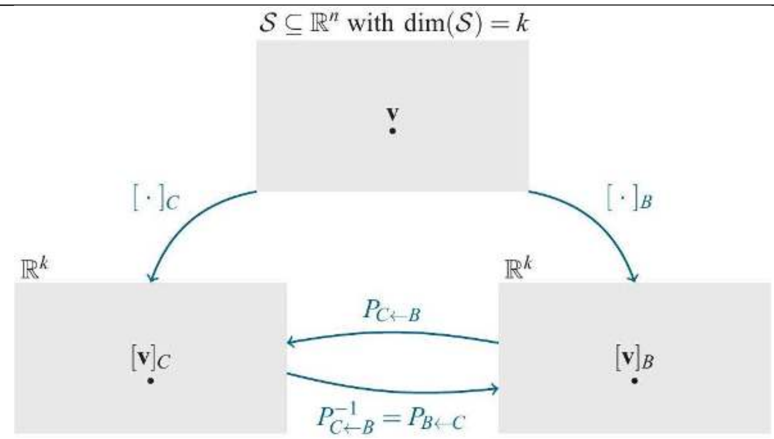
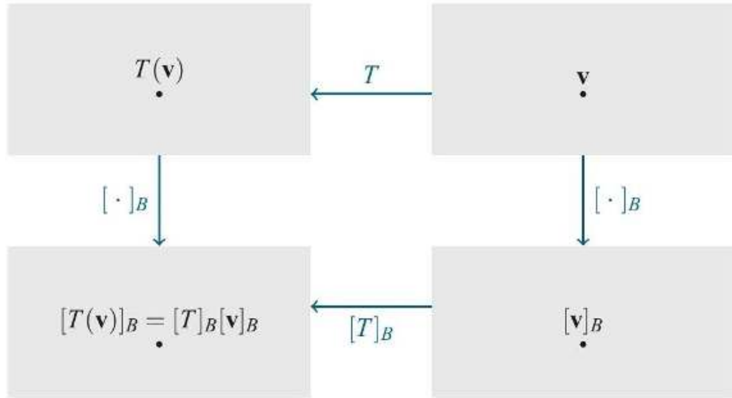
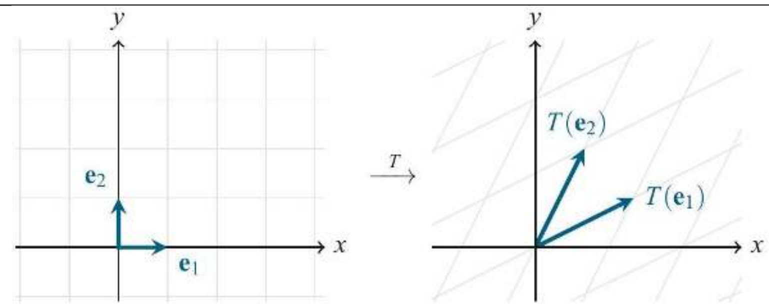
johnston-linear-matrix-algebra_DIA_0130	261	—	—	—	
johnston-linear-matrix-algebra_DIA_0131	263	—	—	—	
johnston-linear-matrix-algebra_DIA_0132	264	—	—	—	

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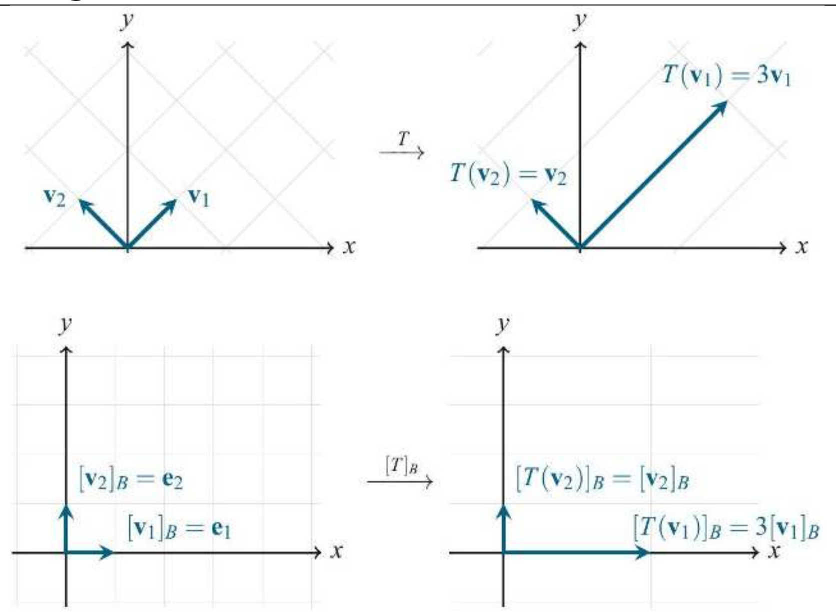
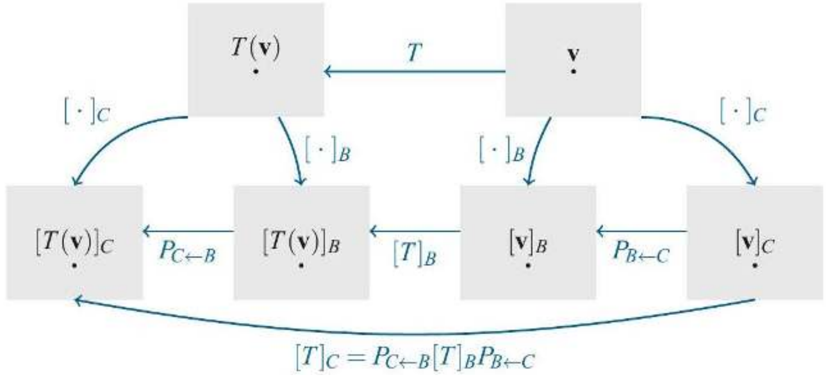
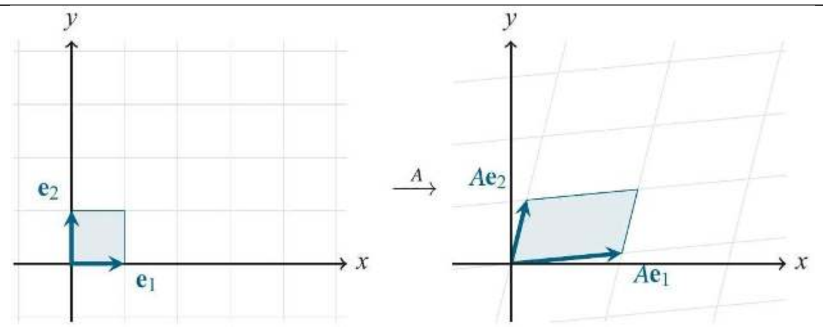
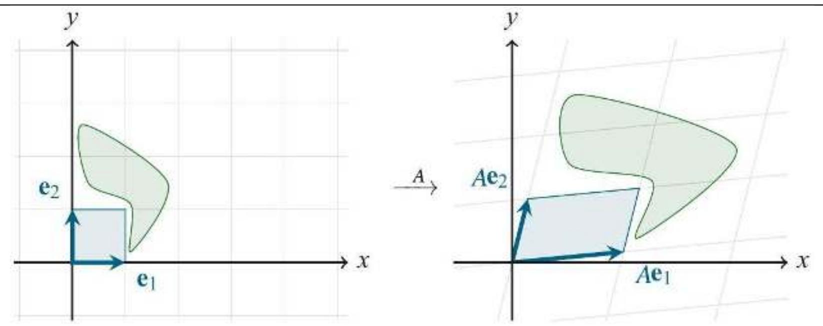
johnston-linear-matrix-algebra_DIA_0133	264	—	—	—	
johnston-linear-matrix-algebra_DIA_0134	265	—	—	—	
johnston-linear-matrix-algebra_DIA_0135	272	—	—	—	
johnston-linear-matrix-algebra_DIA_0136	273	—	—	—	

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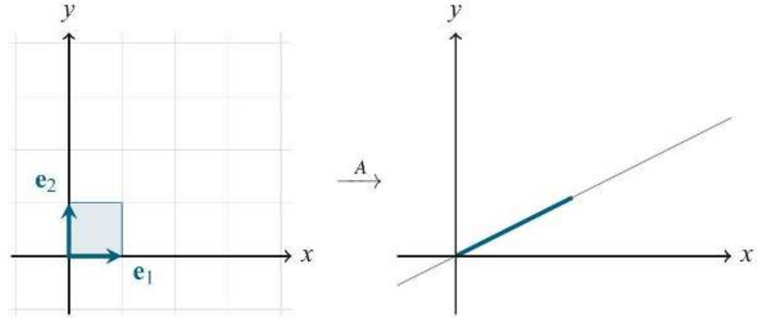
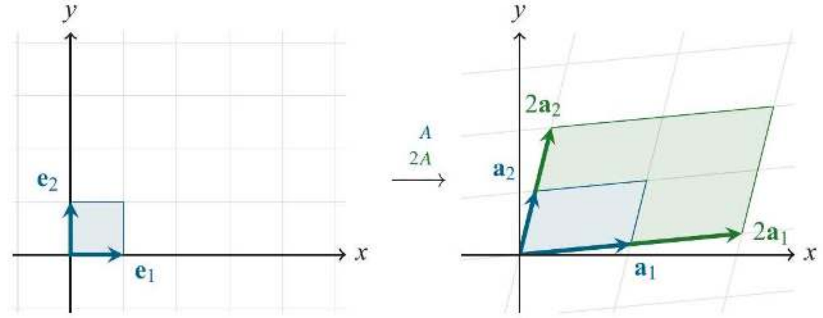
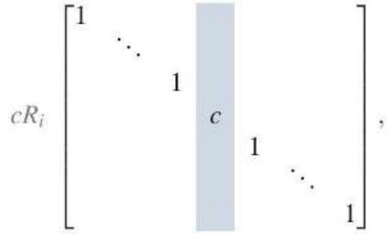
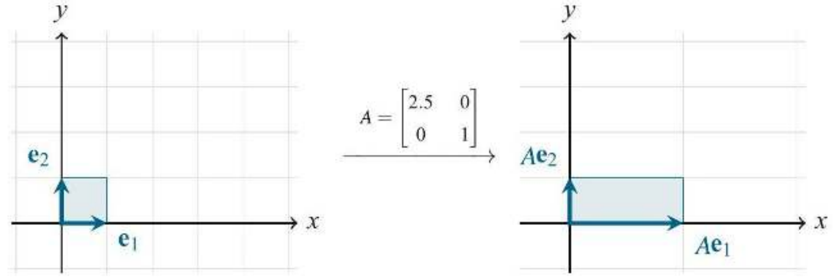
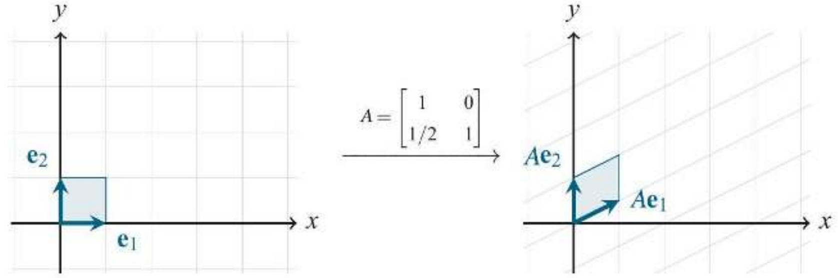
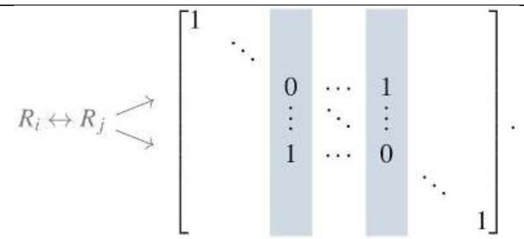
johnston-linear-matrix-algebra_DIA_0137	275	—	—	—	
johnston-linear-matrix-algebra_DIA_0138	276	—	—	—	
johnston-linear-matrix-algebra_DIA_0139	277	—	—	—	
johnston-linear-matrix-algebra_DIA_0140	278	—	—	—	
johnston-linear-matrix-algebra_DIA_0141	278	—	—	—	
johnston-linear-matrix-algebra_DIA_0142	279	—	—	—	

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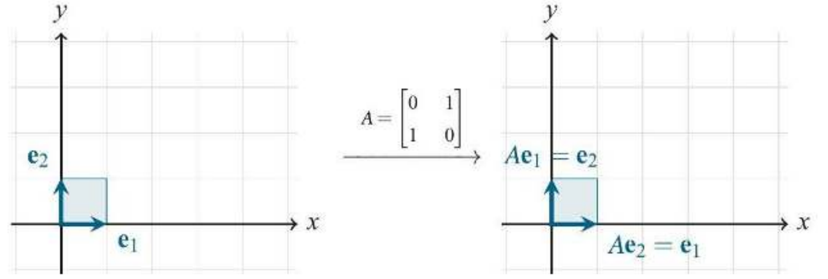
johnston-linear-matrix-algebra_DIA_0143	280	—	—	—	
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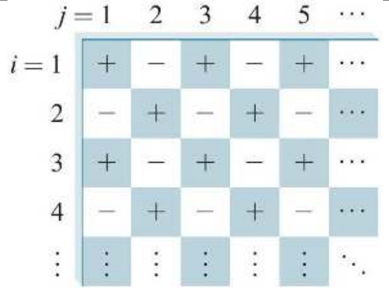
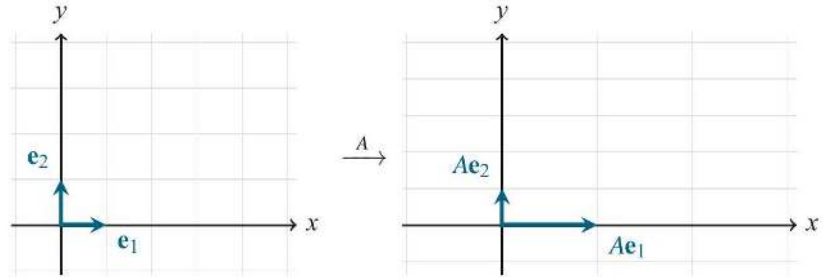
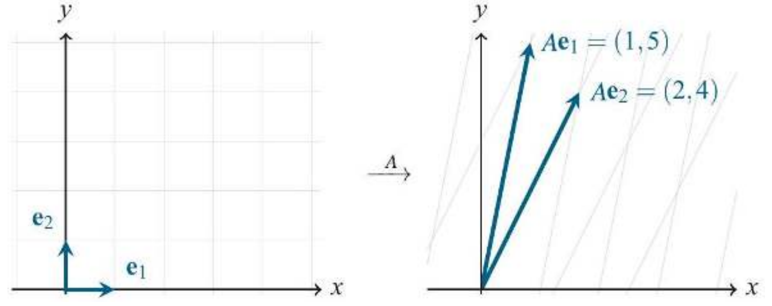
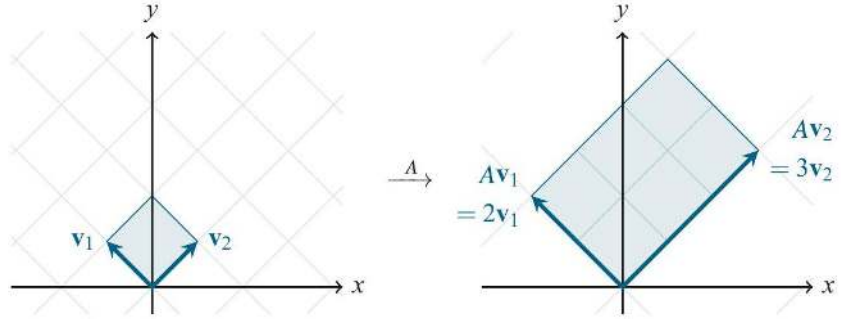
johnston-linear-matrix-algebra_DIA_0145	289	—	—	—	
johnston-linear-matrix-algebra_DIA_0146	299	—	—	—	
johnston-linear-matrix-algebra_DIA_0147	303	—	—	—	
johnston-linear-matrix-algebra_DIA_0148	312	—	—	—	

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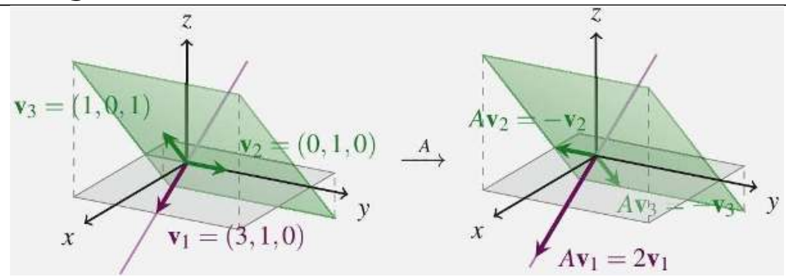
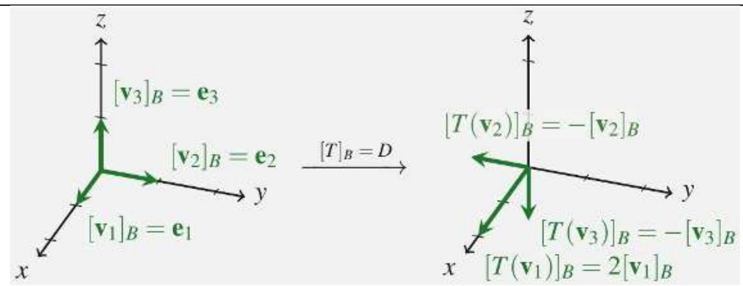
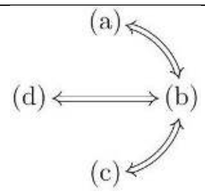
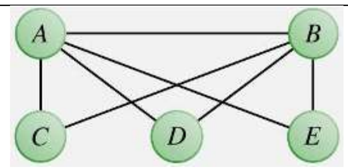
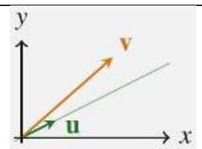
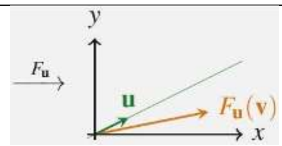
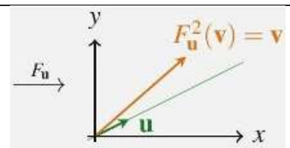
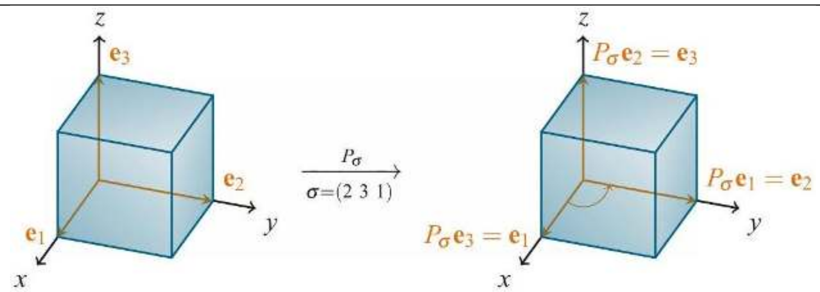
johnston-linear-matrix-algebra_DIA_0149	317	—	—	—	
johnston-linear-matrix-algebra_DIA_0150	328	—	—	—	
johnston-linear-matrix-algebra_DIA_0151	331	—	—	—	
johnston-linear-matrix-algebra_DIA_0152	339	—	—	—	
johnston-linear-matrix-algebra_DIA_0153	344	—	—	—	
johnston-linear-matrix-algebra_DIA_0154	344	—	—	—	
johnston-linear-matrix-algebra_DIA_0155	344	—	—	—	
johnston-linear-matrix-algebra_DIA_0156	363	—	—	—	

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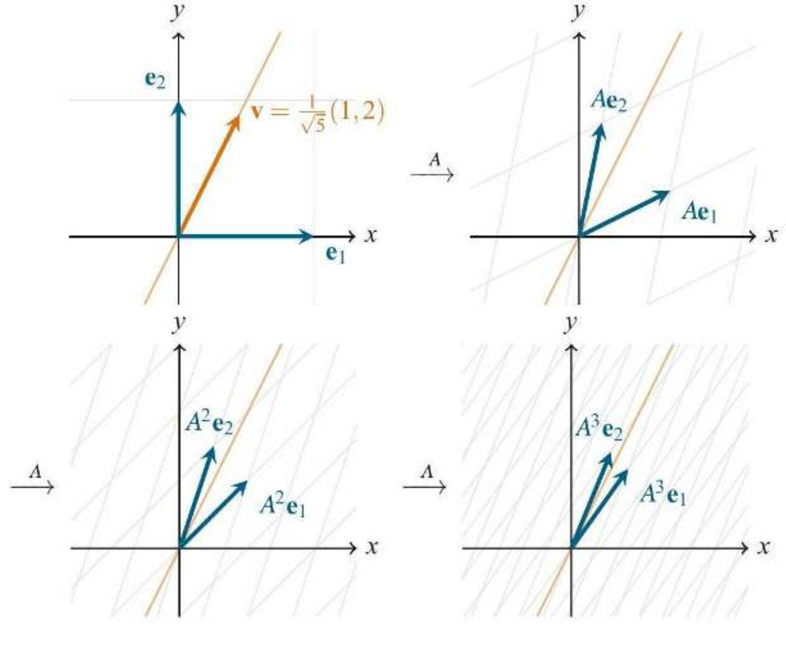
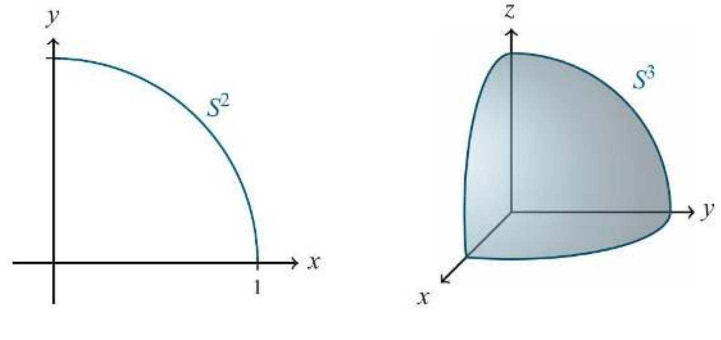
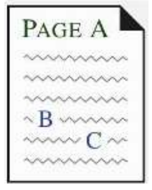
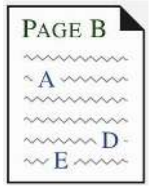
johnston-linear-matrix-algebra_DIA_0157	371	—	—	—	
johnston-linear-matrix-algebra_DIA_0158	382	—	—	—	
johnston-linear-matrix-algebra_DIA_0159	386	—	—	—	
johnston-linear-matrix-algebra_DIA_0160	386	—	—	—	
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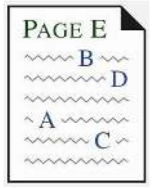
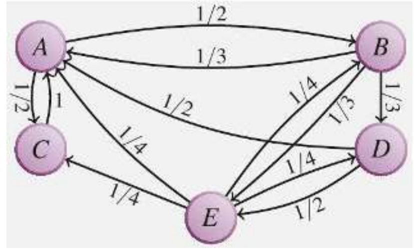
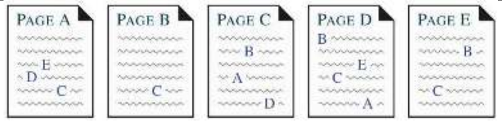
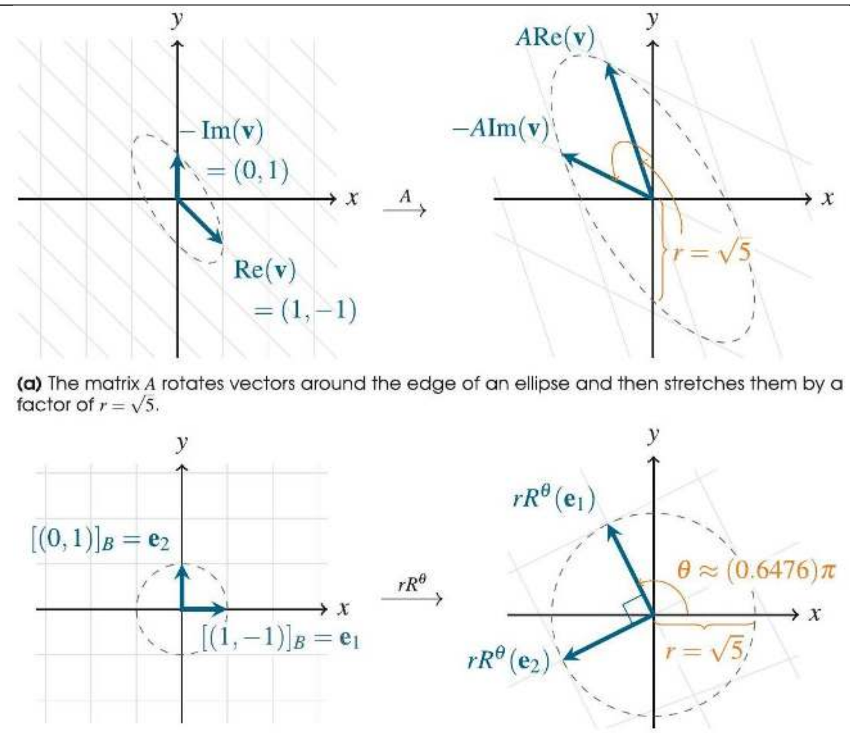
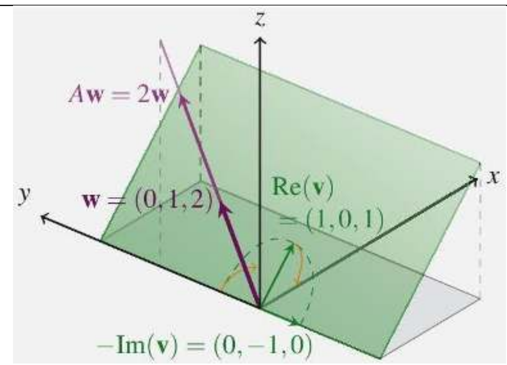
johnston-linear-matrix-algebra_DIA_0163	386	—	—	—	
johnston-linear-matrix-algebra_DIA_0164	387	—	—	—	
johnston-linear-matrix-algebra_DIA_0165	390	—	—	—	
johnston-linear-matrix-algebra_DIA_0166	393	—	—	—	 **(a)** The matrix A rotates vectors around the edge of an ellipse and then stretches them by a factor of $r = \sqrt{5}$. **(b)** In the basis $B = \{\mathrm{Re}(\mathbf{v}), -\mathrm{Im}(\mathbf{v})\} = \{(1,-1),(0,1)\}$, the matrix A can be seen as rotating counter-clockwise by an angle of $\theta \approx (0.6476)\pi$ and stretching by a factor of $r = \sqrt{5}$.
johnston-linear-matrix-algebra_DIA_0167	398	—	—	—	

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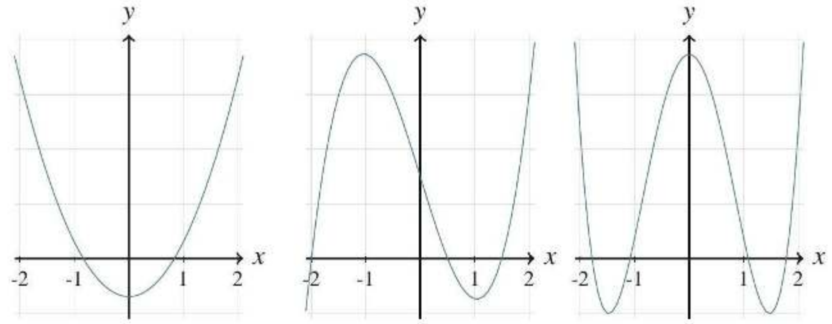
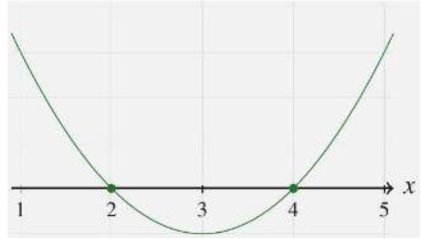
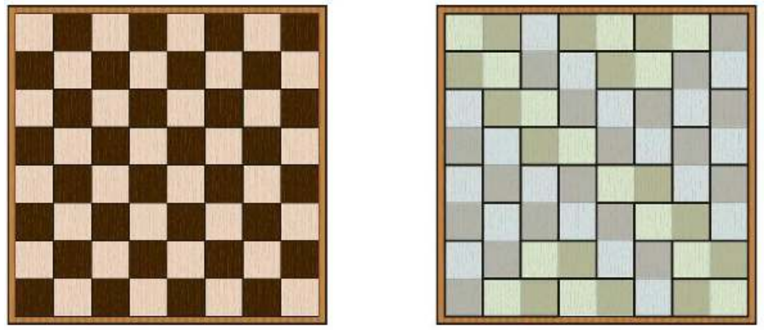
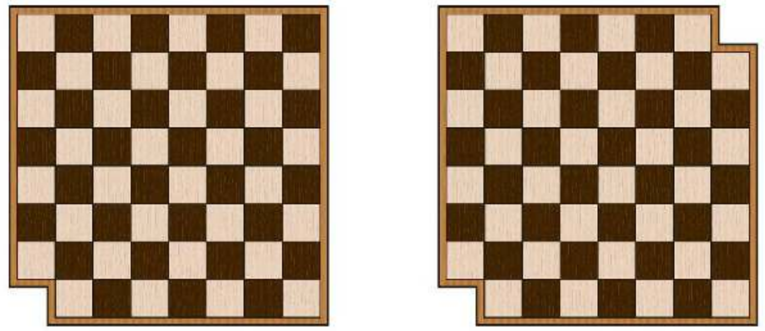
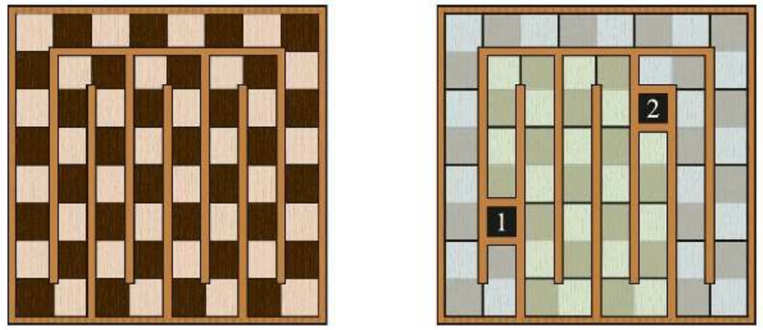
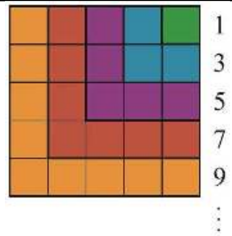
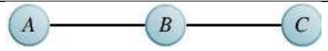
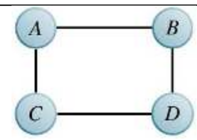
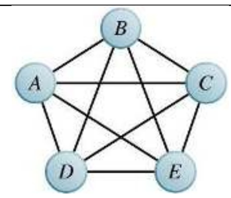
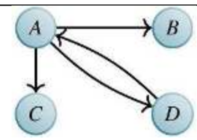
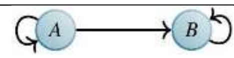
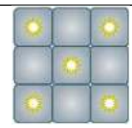
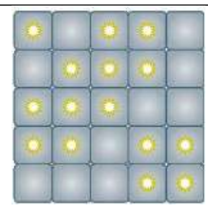
johnston-linear-matrix-algebra_DIA_0168	422	—	—	—	 **(a)** A quadratic (degree 2). **(b)** A cubic (degree 3). **(c)** A quartic (degree 4).
johnston-linear-matrix-algebra_DIA_0169	423	—	—	—	
johnston-linear-matrix-algebra_DIA_0170	432	—	—	—	 **(a)** A standard 8×8 chessboard. **(b)** A domino tiling of a chessboard.
johnston-linear-matrix-algebra_DIA_0171	432	—	—	—	 **(a)** Any chessboard with one square removed cannot be tiled by dominoes. **(b)** This board with two opposite corners removed cannot be tiled by dominoes.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_DIA_0172	433	—	—	—	 **(a)** An arrangement of "walls" on a chess-board that will help us tile it. **(b)** A tiling of a board with (opposite-colored) squares "1" and "2" removed.
johnston-linear-matrix-algebra_DIA_0173	436	—	—	—	
johnston-linear-matrix-algebra_DIA_0174	454	—	—	—	
johnston-linear-matrix-algebra_DIA_0175	454	—	—	—	
johnston-linear-matrix-algebra_DIA_0176	455	—	—	—	
johnston-linear-matrix-algebra_DIA_0177	455	—	—	—	
johnston-linear-matrix-algebra_DIA_0178	455	—	—	—	
johnston-linear-matrix-algebra_DIA_0179	467	—	—	—	
johnston-linear-matrix-algebra_DIA_0180	467	—	—	—	

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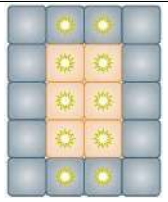
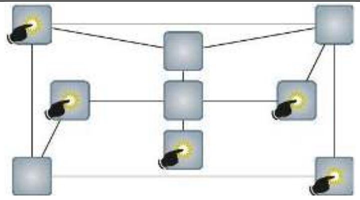
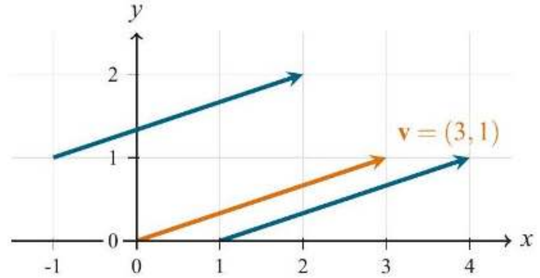
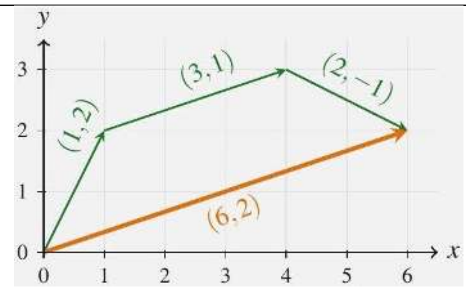
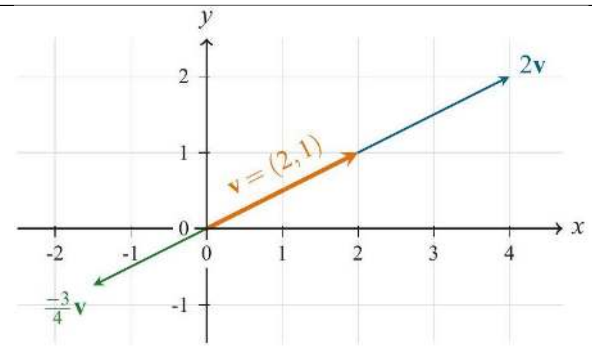
johnston-linear-matrix-algebra_DIA_0182	468	—	—	—	
johnston-linear-matrix-algebra_DIA_0183	468	—	—	—	$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra_DIA_0184	468	—	—	—	
johnston-linear-matrix-algebra_PIC_0001	17	—	—	—	
johnston-linear-matrix-algebra_PIC_0002	19	—	—	—	
johnston-linear-matrix-algebra_PIC_0003	20	—	—	—	

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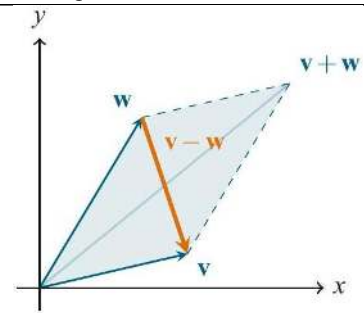
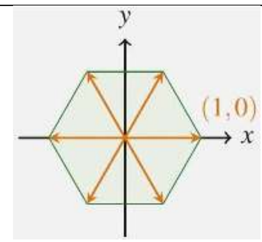
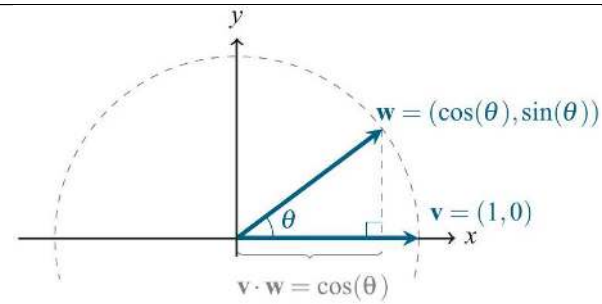
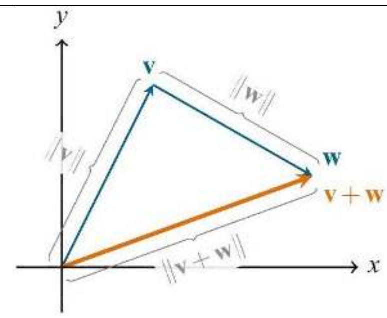
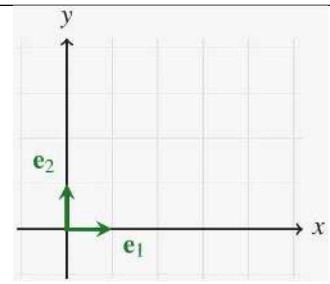
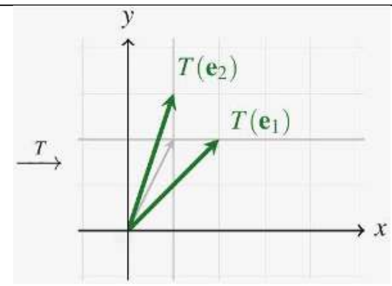
johnston-linear-matrix-algebra_PIC_0004	21	—	—	—	
johnston-linear-matrix-algebra_PIC_0005	22	—	—	—	
johnston-linear-matrix-algebra_PIC_0006	26	—	—	—	
johnston-linear-matrix-algebra_PIC_0007	31	—	—	—	
johnston-linear-matrix-algebra_PIC_0008	52	—	—	—	
johnston-linear-matrix-algebra_PIC_0009	52	—	—	—	

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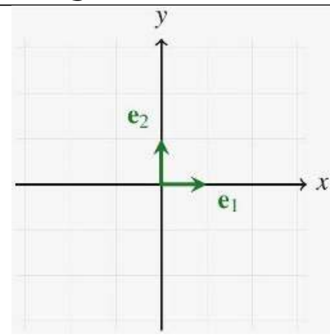
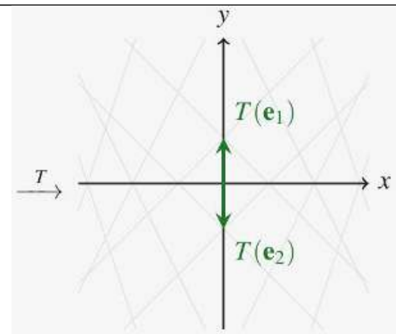
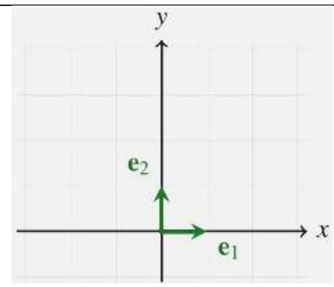
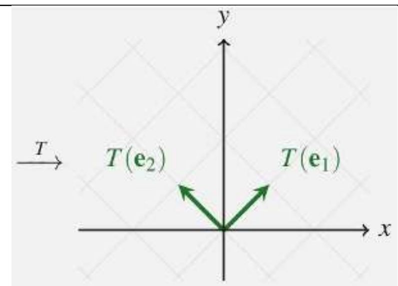
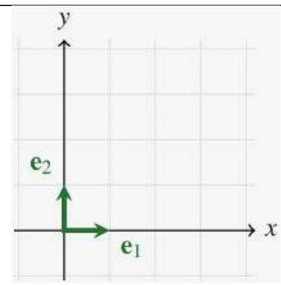
johnston-linear-matrix-algebra_PIC_0010	52	—	—	—	
johnston-linear-matrix-algebra_PIC_0011	52	—	—	—	
johnston-linear-matrix-algebra_PIC_0012	53	—	—	—	
johnston-linear-matrix-algebra_PIC_0013	53	—	—	—	
johnston-linear-matrix-algebra_PIC_0014	55	—	—	—	

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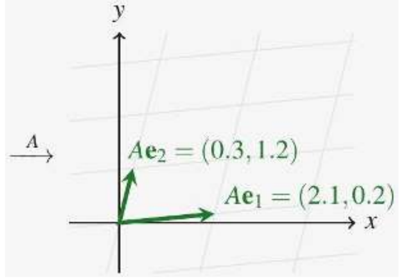
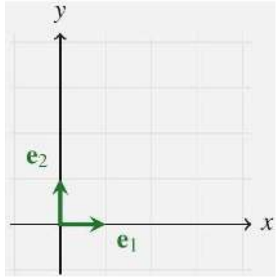
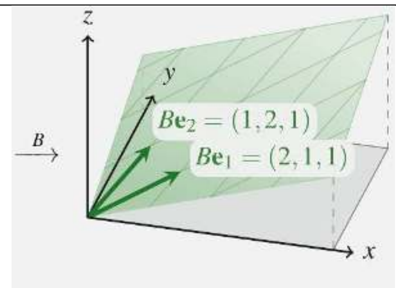
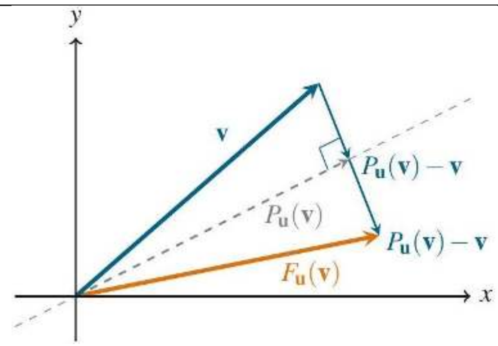
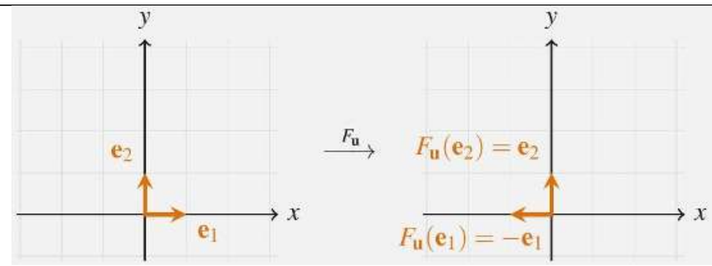
johnston-linear-matrix-algebra_PIC_0015	55	—	—	—	
johnston-linear-matrix-algebra_PIC_0016	56	—	—	—	
johnston-linear-matrix-algebra_PIC_0017	56	—	—	—	
johnston-linear-matrix-algebra_PIC_0018	60	—	—	—	
johnston-linear-matrix-algebra_PIC_0019	60	—	—	—	

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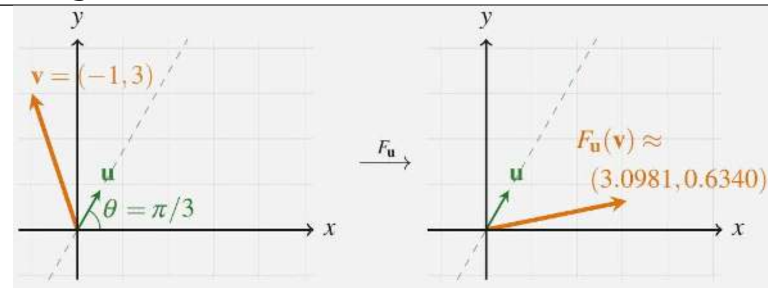
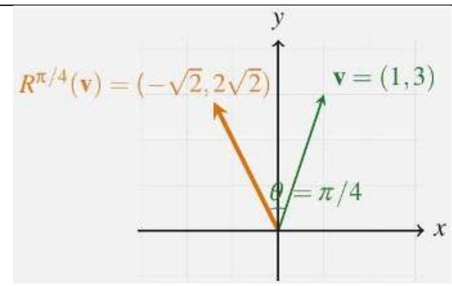
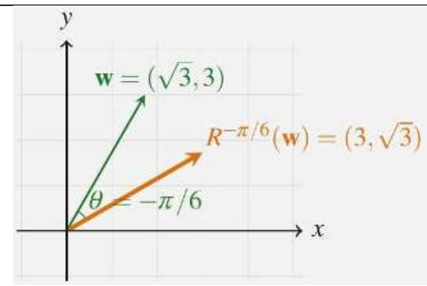
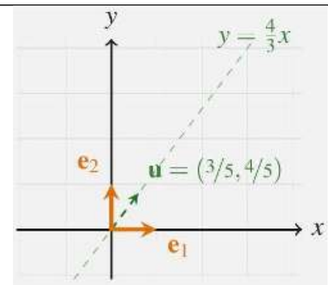
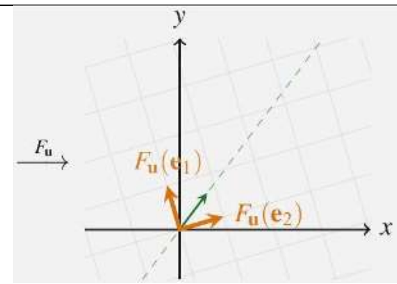
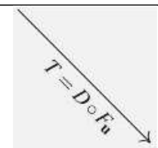
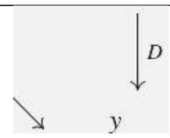
johnston-linear-matrix-algebra_PIC_0020	62	—	—	—	
johnston-linear-matrix-algebra_PIC_0021	64	—	—	—	
johnston-linear-matrix-algebra_PIC_0022	64	—	—	—	
johnston-linear-matrix-algebra_PIC_0023	68	—	—	—	
johnston-linear-matrix-algebra_PIC_0024	68	—	—	—	
johnston-linear-matrix-algebra_PIC_0025	68	—	—	—	
johnston-linear-matrix-algebra_PIC_0026	68	—	—	—	

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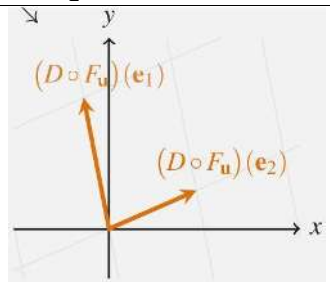
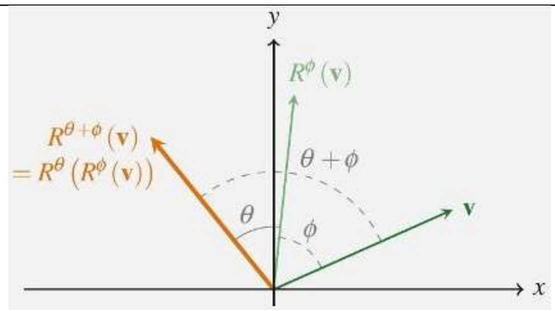
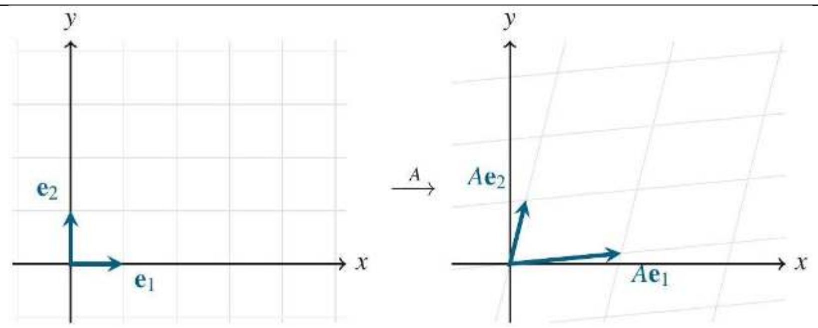
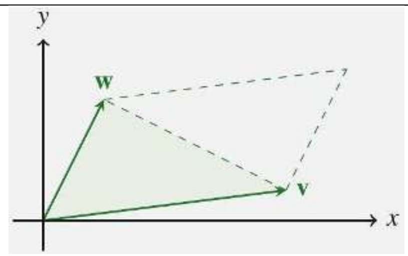
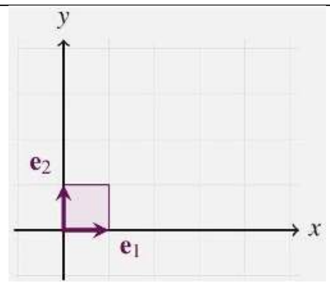
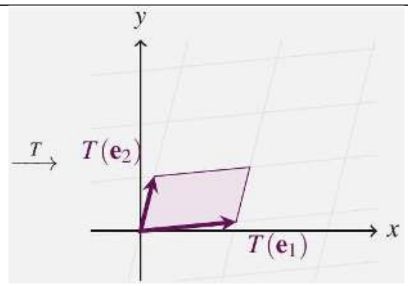
johnston-linear-matrix-algebra_PIC_0027	68	—	—	—	
johnston-linear-matrix-algebra_PIC_0028	69	—	—	—	
johnston-linear-matrix-algebra_PIC_0029	72	—	—	—	
johnston-linear-matrix-algebra_PIC_0030	77	—	—	—	
johnston-linear-matrix-algebra_PIC_0031	79	—	—	—	
johnston-linear-matrix-algebra_PIC_0032	79	—	—	—	

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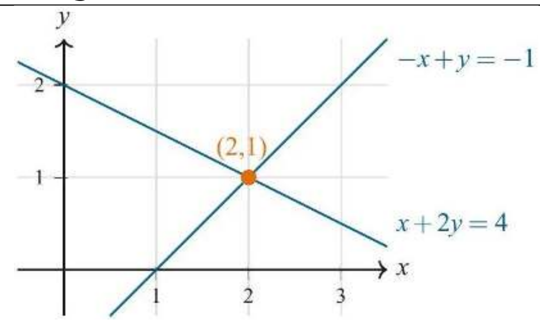
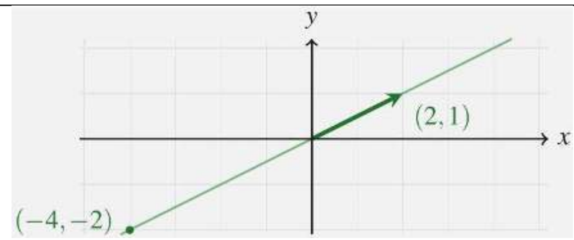
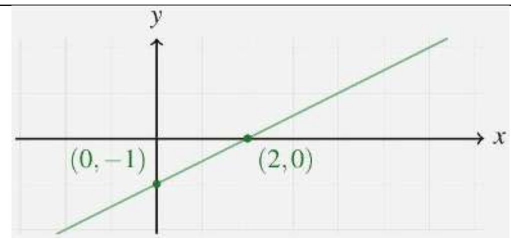
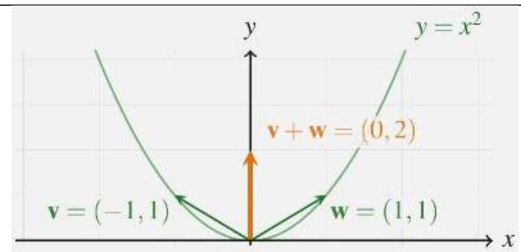
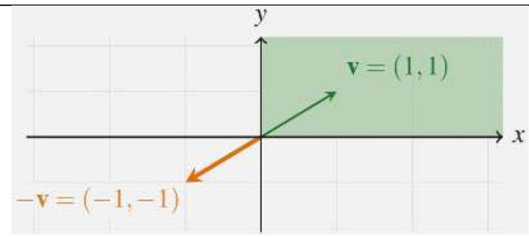
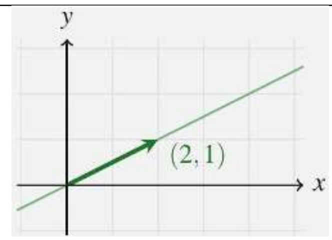
johnston-linear-matrix-algebra_PIC_0033	94	—	—	—	
johnston-linear-matrix-algebra_PIC_0034	137	—	—	—	
johnston-linear-matrix-algebra_PIC_0035	137	—	—	—	
johnston-linear-matrix-algebra_PIC_0036	138	—	—	—	
johnston-linear-matrix-algebra_PIC_0037	139	—	—	—	
johnston-linear-matrix-algebra_PIC_0038	140	—	—	—	

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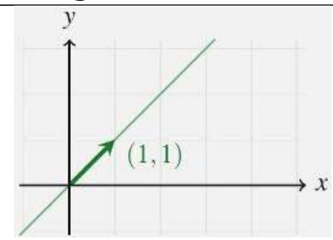
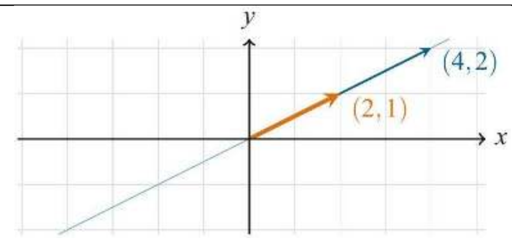
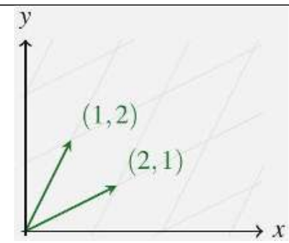
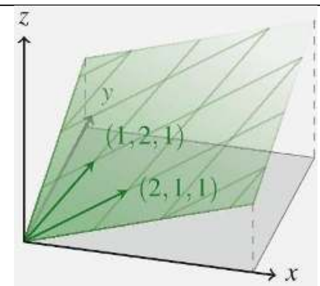
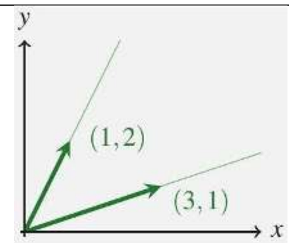
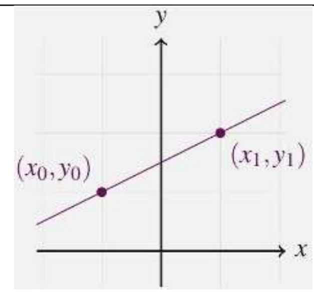
johnston-linear-matrix-algebra_PIC_0039	140	—	—	—	
johnston-linear-matrix-algebra_PIC_0040	141	—	—	—	
johnston-linear-matrix-algebra_PIC_0041	142	—	—	—	
johnston-linear-matrix-algebra_PIC_0042	143	—	—	—	
johnston-linear-matrix-algebra_PIC_0043	149	—	—	—	
johnston-linear-matrix-algebra_PIC_0044	152	—	—	—	

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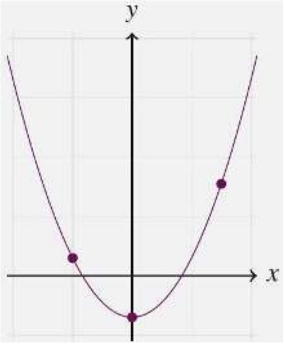
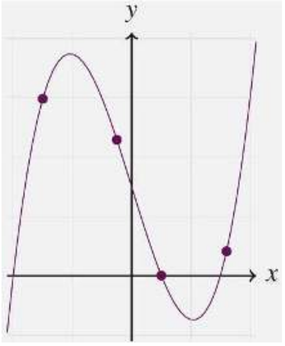
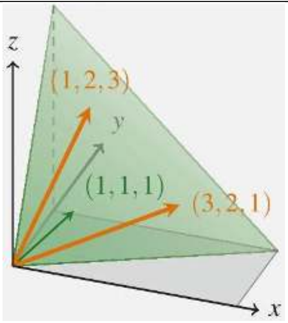
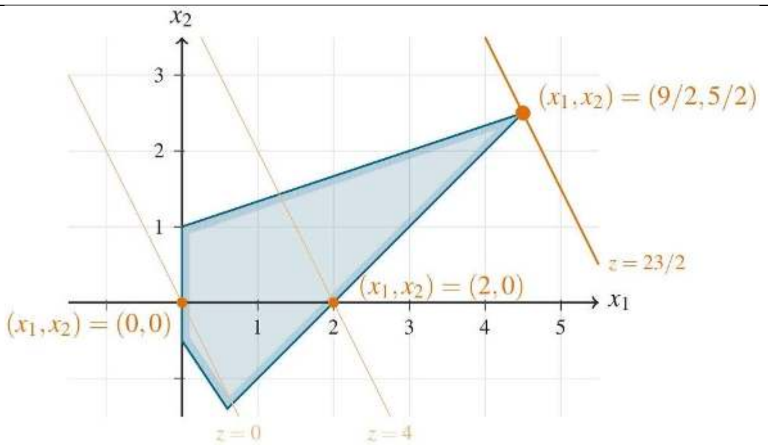
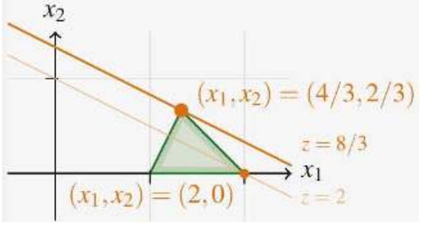
johnston-linear-matrix-algebra_PIC_0045	152	—	—	—	
johnston-linear-matrix-algebra_PIC_0046	152	—	—	—	
johnston-linear-matrix-algebra_PIC_0047	160	—	—	—	
johnston-linear-matrix-algebra_PIC_0048	208	—	—	—	
johnston-linear-matrix-algebra_PIC_0049	214	—	—	—	

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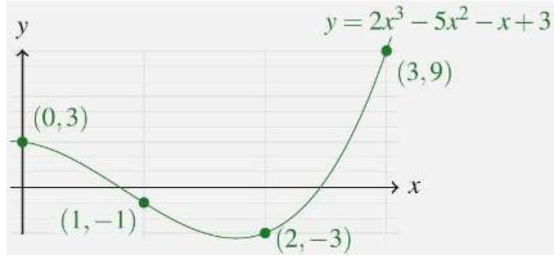
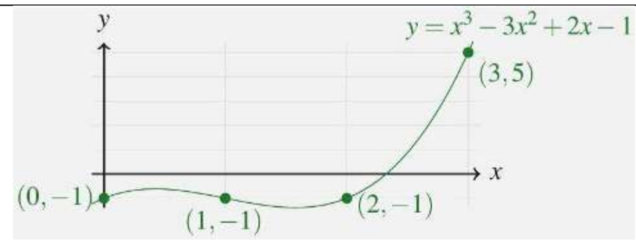
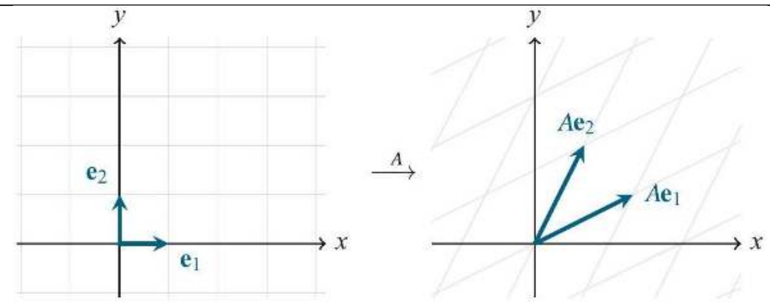
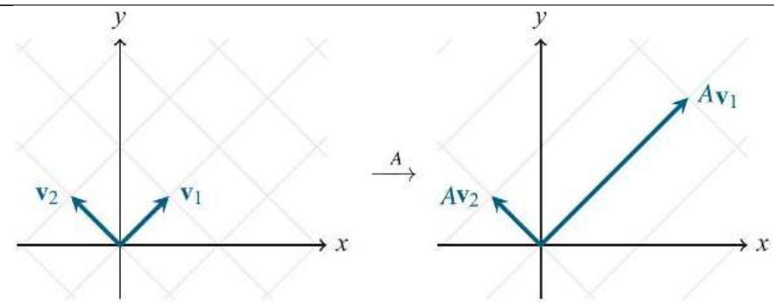
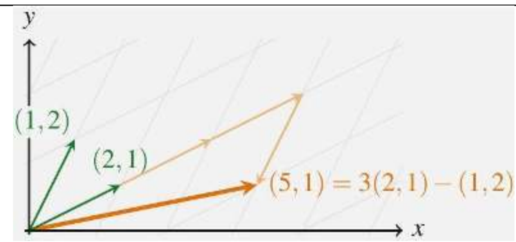
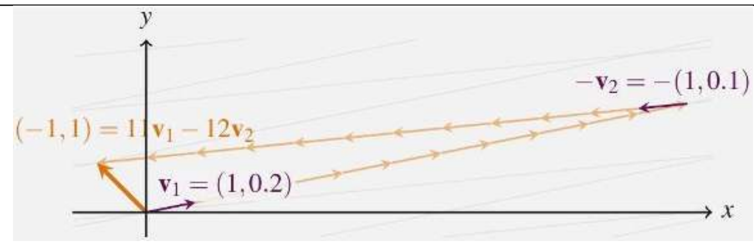
johnston-linear-matrix-algebra_PIC_0050	239	—	—	—	
johnston-linear-matrix-algebra_PIC_0051	240	—	—	—	
johnston-linear-matrix-algebra_PIC_0052	249	—	—	—	
johnston-linear-matrix-algebra_PIC_0053	250	—	—	—	
johnston-linear-matrix-algebra_PIC_0054	253	—	—	—	
johnston-linear-matrix-algebra_PIC_0055	256	—	—	—	

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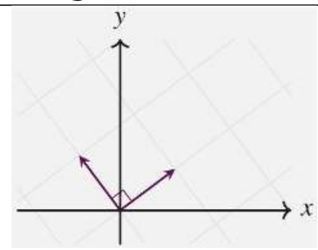
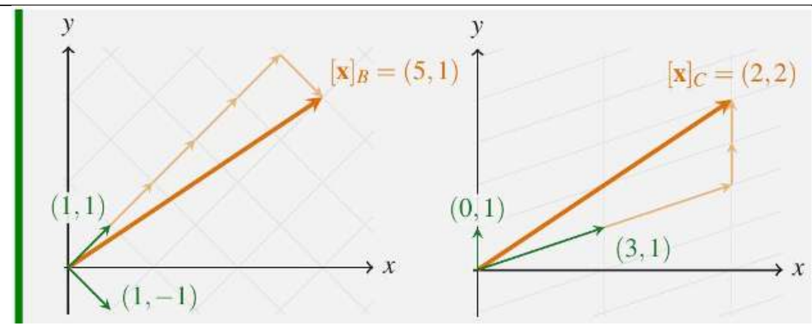
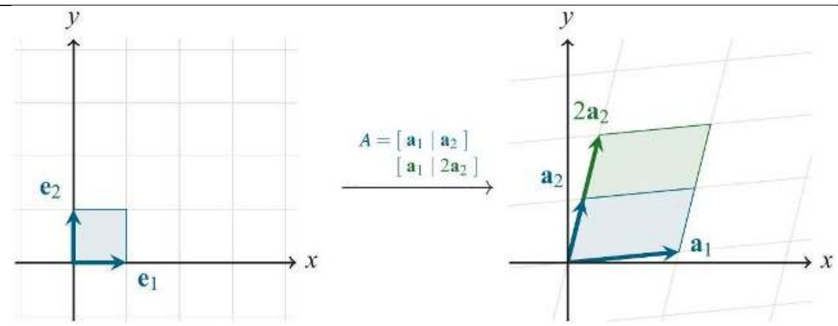
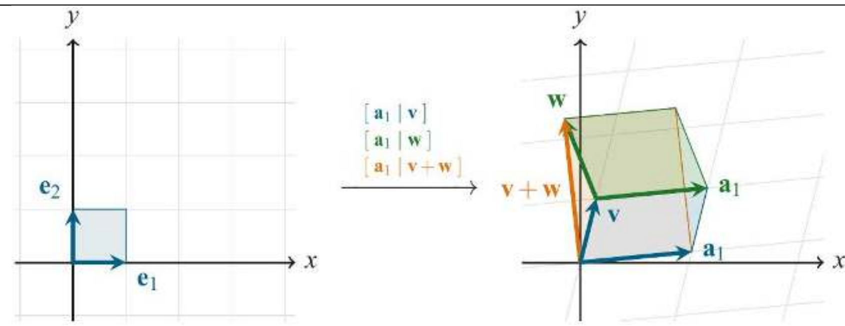
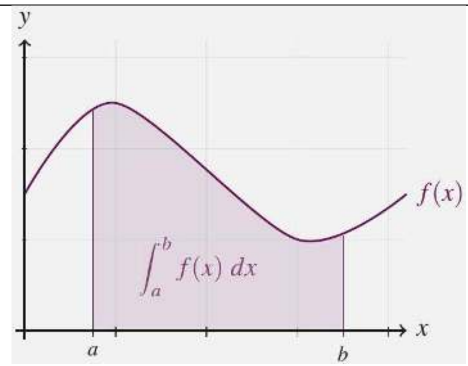
johnston-linear-matrix-algebra_PIC_0056	257	—	—	—	
johnston-linear-matrix-algebra_PIC_0057	258	—	—	—	
johnston-linear-matrix-algebra_PIC_0058	259	—	—	—	$\mathbf{w}_1 = (1,-1,1,0,2), \quad \mathbf{w}_2 = (1,1,2,0,-1), \quad \mathbf{w}_3 = (2,3,1,1,-1).$
johnston-linear-matrix-algebra_PIC_0059	274	—	—	—	
johnston-linear-matrix-algebra_PIC_0060	274	—	—	—	
johnston-linear-matrix-algebra_PIC_0061	280	—	—	—	

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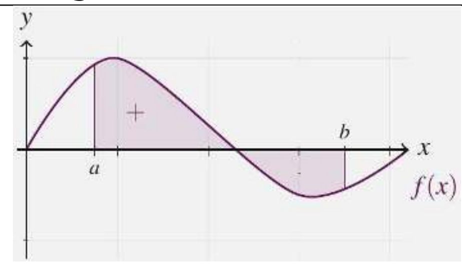
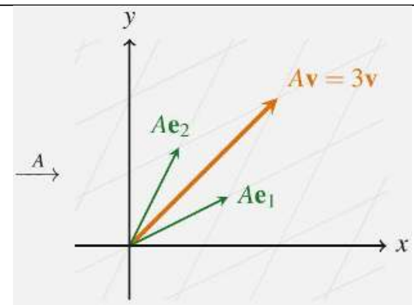
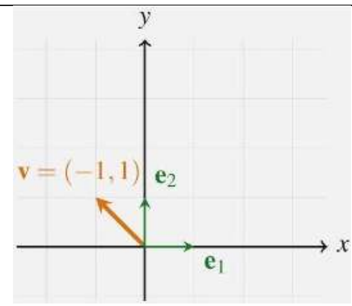
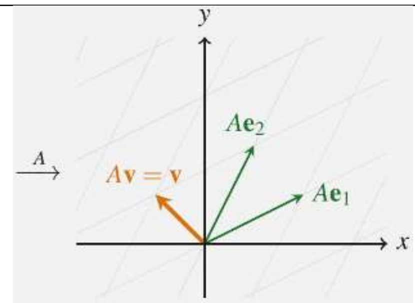
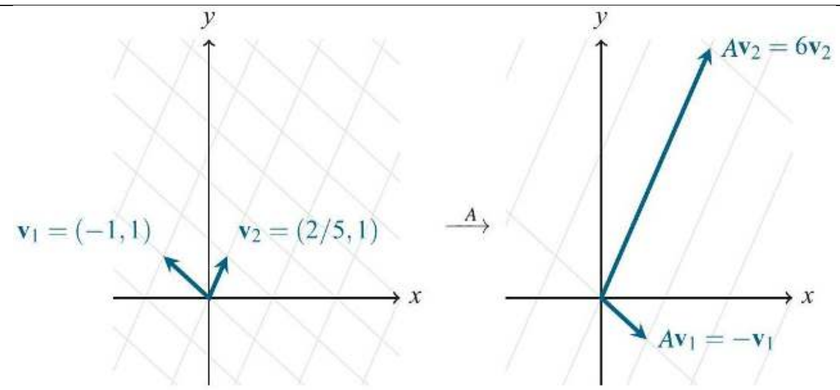
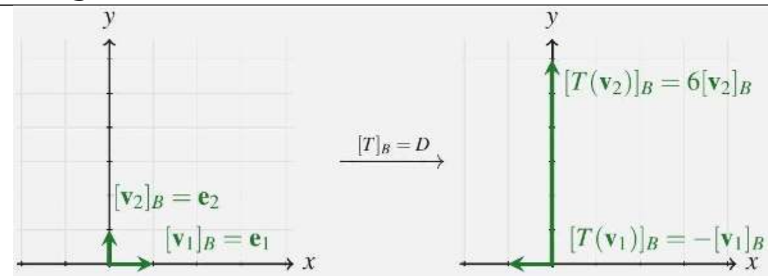
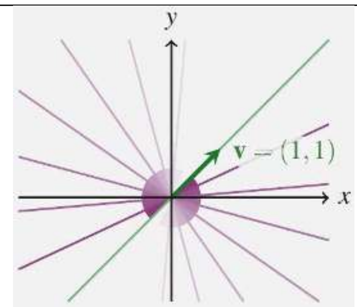
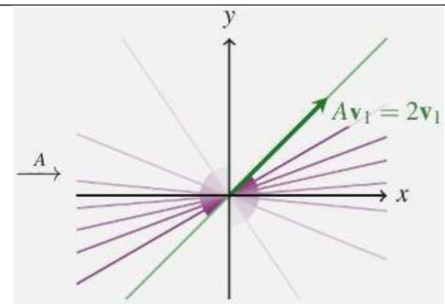
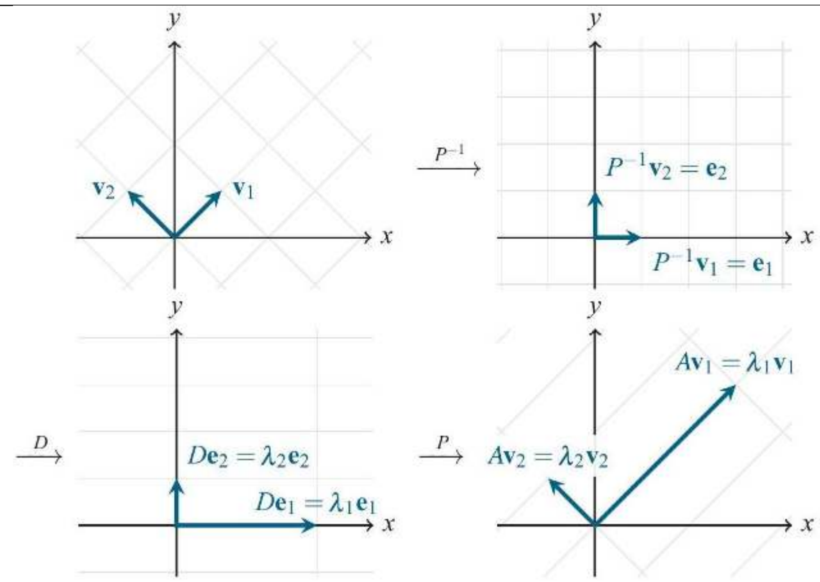
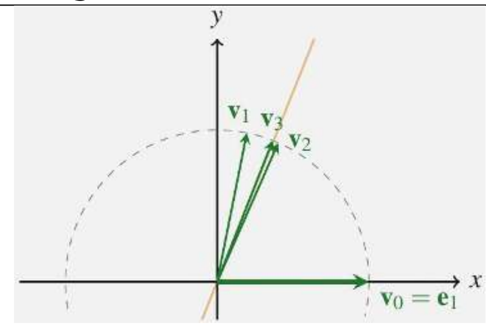
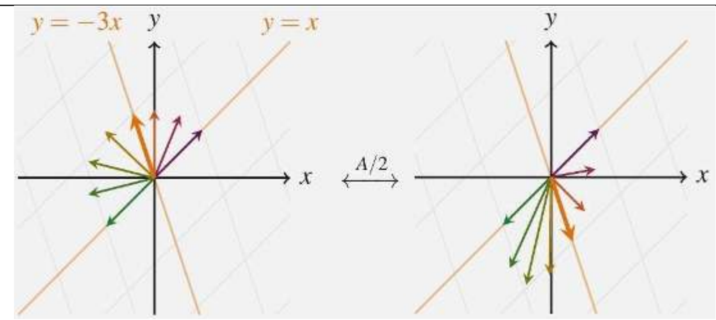
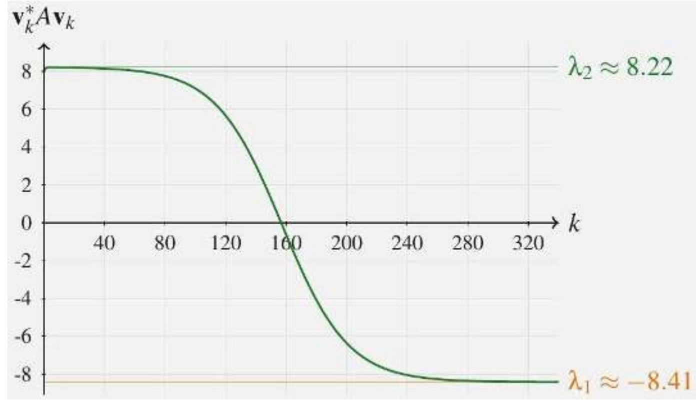
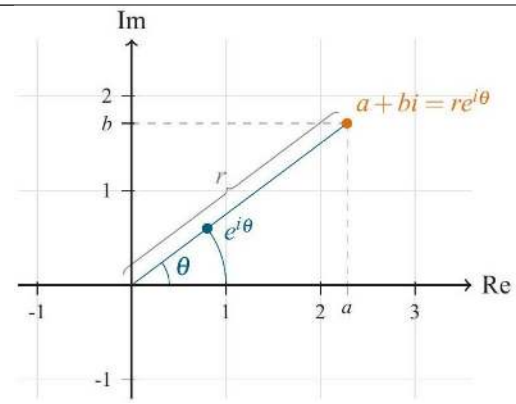
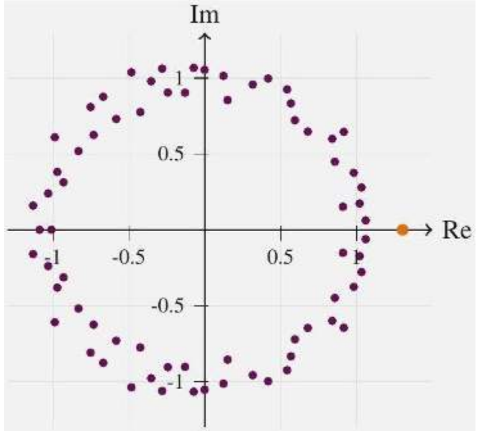
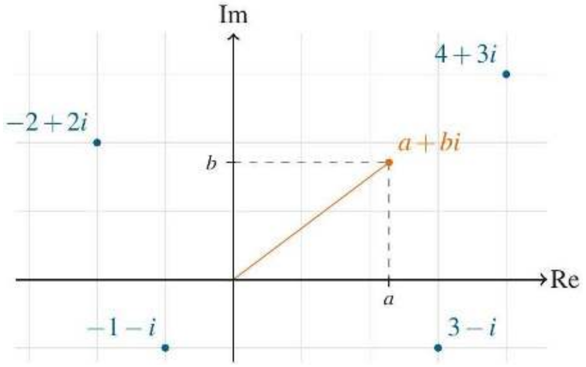
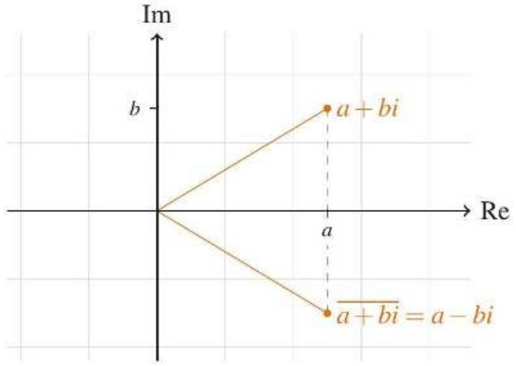
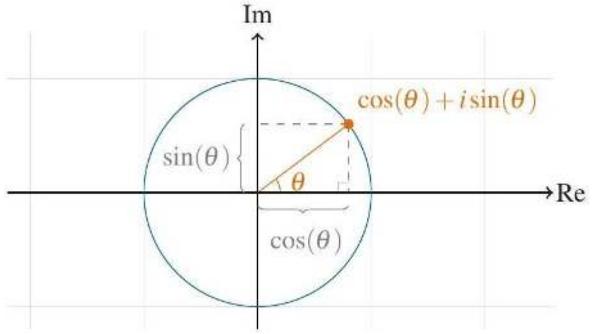
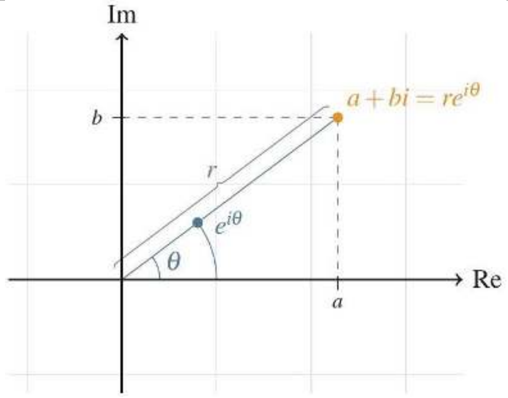
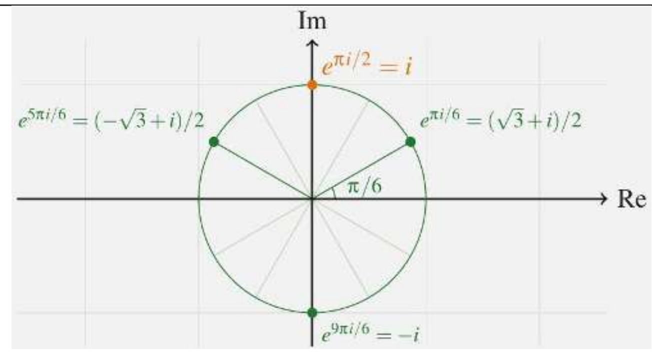
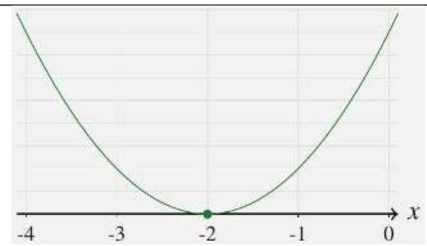
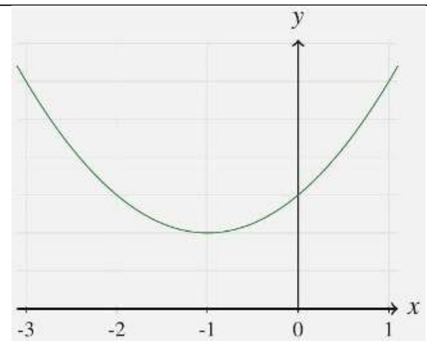
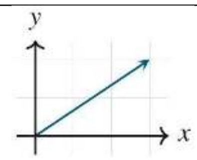
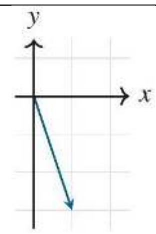
johnston-linear-matrix-algebra_PIC_0062	281	—	—	—	
johnston-linear-matrix-algebra_PIC_0063	300	—	—	—	
johnston-linear-matrix-algebra_PIC_0064	301	—	—	—	
johnston-linear-matrix-algebra_PIC_0065	301	—	—	—	
johnston-linear-matrix-algebra_PIC_0066	304	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_PIC_0067	327	—	—	—	
johnston-linear-matrix-algebra_PIC_0068	331	—	—	—	
johnston-linear-matrix-algebra_PIC_0069	331	—	—	—	
johnston-linear-matrix-algebra_PIC_0070	352	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_PIC_0071	372	—	—	—	
johnston-linear-matrix-algebra_PIC_0072	376	—	—	—	
johnston-linear-matrix-algebra_PIC_0073	379	—	—	—	
johnston-linear-matrix-algebra_PIC_0074	392	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_PIC_0075	414	—	—	—	
johnston-linear-matrix-algebra_PIC_0076	418	—	—	—	
johnston-linear-matrix-algebra_PIC_0077	418	—	—	—	
johnston-linear-matrix-algebra_PIC_0078	420	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_ PIC_0079	421	—	—	—	
johnston-linear-matrix-algebra_ PIC_0080	421	—	—	—	
johnston-linear-matrix-algebra_ PIC_0081	423	—	—	—	
johnston-linear-matrix-algebra_ PIC_0082	424	—	—	—	
johnston-linear-matrix-algebra_ PIC_0083	446	—	—	—	
johnston-linear-matrix-algebra_ PIC_0084	446	—	—	—	

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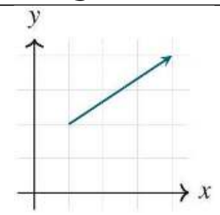
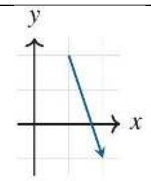
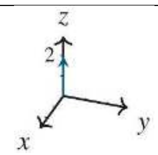
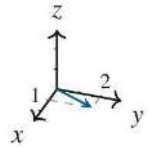
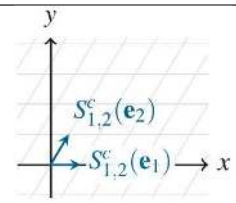
johnston-linear-matrix-algebra_PIC_0085	446	—	—	—	
johnston-linear-matrix-algebra_PIC_0086	446	—	—	—	
johnston-linear-matrix-algebra_PIC_0087	446	—	—	—	
johnston-linear-matrix-algebra_PIC_0088	446	—	—	—	
johnston-linear-matrix-algebra_PIC_0089	452	—	—	—	
johnston-linear-matrix-algebra_PIC_0090	452	—	—	—